A REALIZATION OF G_2 FOR LITT PROBLEM NO. 13

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ABSTRACT. We exhibit a nonempty family of complex abelian threefolds P with integral symmetric ample divisors $X \subset P$ for which the full convolution Tannaka group is the simple group of type G_2. The family comes from cyclic triple covers $C \to E$ of elliptic curves, written as $C : w^6 = f_3(x)$. The induced Prym polarization has type $(1, 1, 3)$. Each theta section X is normal and has one simple elliptic singularity of type $\widetilde{E}_7$; its intersection complex $K = \mathrm{IC}_X$ satisfies

$$\chi(K) = 7, \qquad K \text{ is a direct summand of } \mathrm{Alt}^2(K).$$

The second assertion is in the convolution Tannakian category. We prove it by spreading two components of a generic pair-incidence curve and extracting the resulting divisor-supported summand. Their calculation rests on an exact norm–conic identity over a rational function field. Finally, the Hodge cocharacter has weight multiplicities $(2, 3, 2)$; together with the exterior-square summand, this identifies the full group with G_2 on its standard seven-dimensional module. Thus Litt Problem No. 13 has an affirmative answer for G_2.

1. Introduction

Let A be a complex abelian variety. Convolution of constructible complexes on A induces a neutral Tannakian structure after quotienting by the negligible objects. Thus an integral subvariety $Z \subset A$ with nonzero intersection-cohomology Euler characteristic determines an algebraic group $G(\mathrm{IC}_Z)$ and a faithful representation of dimension $\chi(\mathrm{IC}_Z)$; see [15]. More precisely, Litt Problem No. 13 asks whether, for a simple algebraic group H of one of the types

$$G_2, \qquad E_7, \qquad E_8, \qquad F_4$$

there exist an abelian variety A and a subvariety $X \subset A$ with $G(\mathrm{IC}_X) \simeq H$ [18]. The theorem below settles precisely the $H = G_2$ instance. It does not address the E_7, E_8, or F_4 instances.

We give a construction for the G_2 case. Let

$$\mathcal{B} = \operatorname{Spec} \mathbf{C}[A, B, (4A^3 + 27B^2)^{-1}].$$

For $b = (A, B) \in \mathcal{B}(\mathbf{C})$, consider the smooth projective curves

$$E_b : y^2 = x^3 + Ax + B, \qquad C_b : w^6 = x^3 + Ax + B,$$

and the cyclic triple covering

$$\pi_b : C_b \longrightarrow E_b, \qquad (x, w) \longmapsto (x, w^3).$$

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Write $O \in E_b$ for the point at infinity, put

$$\kappa_b = \pi_b^* \mathcal{O}_{E_b}(O), \qquad P_b = \ker(\mathrm{Nm}_{\pi_b})^0,$$

and define the cyclic theta section, with the scheme structure cut out by the restriction of the centred theta section,

$$X_b = \{M \in P_b : h^0(C_b, \kappa_b \otimes M) > 0\}.$$

Theorem 1.1 (Main theorem)**.** *There is a nonempty Zariski-open subset $\mathcal{B}^\circ \subset \mathcal{B}$ such that, for every $b \in \mathcal{B}^\circ(\mathbf{C})$, the variety P_b is an abelian threefold, $X_b \subset P_b$ is an integral ample divisor satisfying $[-1]_{P_b}^* X_b = X_b$, and there is an isomorphism of pairs of complex algebraic groups and faithful modules*

$$\big(G(\mathrm{IC}_{X_b}), \omega(\mathrm{IC}_{X_b})\big) \simeq (G_2, V_7),$$

where G_2 is the connected simple complex group of type G_2 and V_7 is its standard seven-dimensional representation. Thus the group on the left is the full convolution Tannaka group.

Proof outline. Fix $b \in \mathcal{B}(\mathbf{C})$ and write $(P, X) = (P_b, X_b)$. The norm fibre in $C^{(3)}$ gives a resolution of X. Its exceptional curves and an Euler-characteristic stratification give

$$\chi(\mathrm{IC}_X) = 7, \qquad \mathrm{Stab}_P(X) = \{0\}.$$

The same resolution, after a generic unitary twist, gives $(h^{2,0}, h^{1,1}, h^{0,2}) = (2, 3, 2)$.

Over the total family, let x be a geometric generic point of the relative smooth theta locus $\mathcal{M} = (\mathcal{X}/\mathcal{B})_{\mathrm{sm}}$, and let D_x be its representing effective divisor of degree three. On the corresponding geometric generic fibre, consider the pair-incidence curve

$$C_x = \{y \in X : -x - y \in X\}.$$

The norm of a section of $H^0(C, 3\kappa - D_x)$ is the fixed norm line times a conic. The determinant of that conic factors as a structural line and an integral residual octic. An exact identity over a rational function field splits the double cover that orders the two line factors. A relative Abel inverse identifies those ordered factors with two components of C_x, and a boundary argument shows that there are no further nonexceptional components.

These two generic components extend over a nonempty open $\mathcal{M}^\circ \subset \mathcal{X} \setminus 0$. Since $\mathcal{M}^\circ \to \mathcal{B}$ is open, its image is a nonempty open $\mathcal{B}^\circ$. For every $b \in \mathcal{B}^\circ(\mathbf{C})$, top cohomology of the pair fibres and the decomposition theorem then give $\mathrm{IC}_X \mid \mathrm{Alt}^2(\mathrm{IC}_X)$ in the convolution Tannakian category.

Writing $G = G(\mathrm{IC}_X)$ and $V = \omega(\mathrm{IC}_X)$, the representation-theoretic inputs are

$$G \subset O(V), \qquad \dim V = 7, \qquad V|_{G^0} \text{ irreducible},$$

$$\mu : \mathbf{G}_m \longrightarrow [G^0, G^0] \quad \text{with weights } -2, 0, 2 \text{ of multiplicities } 2, 3, 2,$$

$$\mathrm{Hom}_G\big(V, \Lambda^2 V\big) \neq 0.$$

They distinguish G_2 from SO_7 and the seven-dimensional PGL_2 module. The same exterior-square morphism rules out the only possible disconnected extension.

Computer-assisted calculations. The passage from a factorization of the residual norm conic to pair incidence is proved scheme-theoretically in Lemma 4.7 and Proposition 4.8. The symbolic calculations derive the residual determinant and verify an identity in $\mathbf{Q}(\lambda, e_2, e_3, u)$. The supplementary scripts check the identity by two algebraically different exact reductions of the same generated formulas. The two finite-field calculations establish, respectively, geometric integrality of one residual-octic fibre and nonemptiness of the admissible locus. The geometric arguments in Lemmas 4.3 and 4.9 then pass to the generic and relative statements used in the proof.

Verification map. For readers checking the argument, the proof has five interfaces, each with a separate output used by the next one.

(1) *Geometry of the theta section:* Proposition 3.3, Corollary 3.4, and Lemma 3.5 prove that X is an integral normal symmetric ample surface with one $\widetilde{E}_7$ singularity, $\chi(\mathrm{IC}_X) = 7$, trivial translation stabilizer, and dominant conormal Gauss map.
(2) *Hodge input:* Lemma 3.6 and Proposition 3.7 compute the generic-twist Hodge multiplicities $(2, 3, 2)$.
(3) *Pair-incidence input:* Propositions 4.2, 4.5 and 4.8 and Lemmas 4.3 and 4.11 identify exactly two nonexceptional components of the geometric generic pair fibre; Theorem 5.1 spreads them to a relative open set.
(4) *Tannakian detection:* Lemmas 6.1 to 6.3 isolate an unshifted IC_X summand in alternating pair convolution, giving Proposition 6.4.
(5) *Group identification:* Proposition 7.1 and Corollary 7.2 combine irreducibility, the Hodge cocharacter, and $\mathrm{Hom}_G(V, \Lambda^2 V) \neq 0$ to identify first G^0 and then the full group with G_2.

The exact scripts are used only at the three explicitly marked algebraic interfaces in (3): determinant derivation, the ordering-cover identity, and two finite-field nonemptiness witnesses. They do not replace the characteristic-zero openness arguments, the decomposition-theorem argument, or the representation-theoretic classification.

All varieties are defined over $\mathbf{C}$. Intersection complexes, cohomology, and Hodge modules retain their canonical $\mathbf{Q}$-structures. For the Tannakian construction we first extend the underlying perverse sheaves from $\mathbf{Q}$ to $\mathbf{C}$, as required by the analytic form of the results cited below. Accordingly, every Tannaka group and tautological module in this paper is over $\mathbf{C}$. We use $\mathbf{P}(\mathcal{E}) = \mathrm{Proj}(\mathrm{Sym}\,\mathcal{E}^\vee)$, so it parametrizes lines in the fibres of $\mathcal{E}$. For an integral scheme Y, we write $k(Y)$ for its function field; for a point y, we write $k(y)$ for its residue field.

2. Tannakian conventions

Let A be a complex abelian variety and let $m : A \times A \to A$ be addition. For constructible complexes with coefficients in $\mathbf{Q}$ or $\mathbf{C}$ we write

$$K * L = Rm_*(K \boxtimes L).$$

Set

$$\mathsf{S}(A) = \{Q \in \mathrm{Perv}(A, \mathbf{C}) : \chi(A, Q) = 0\},$$
$$\mathsf{T}(A) = \{M \in D^b_c(A, \mathbf{C}) : {}^pH^i(M) \in \mathsf{S}(A) \text{ for every } i\}.$$

The first is a Serre subcategory and the second is the corresponding triangulated tensor ideal. The quotient

$$\overline{D}(A) = D^b_c(A, \mathbf{C})/\mathsf{T}(A)$$

inherits the perverse t-structure; its heart is

$$\overline{\mathrm{Perv}}(A) = \mathrm{Perv}(A, \mathbf{C})/\mathsf{S}(A).$$

Convolution is t-exact on $\overline{D}(A)$, and the displayed heart is a rigid symmetric monoidal neutral Tannakian category [11, Proposition 3.1]. For the pure polarizable objects used below, the tensor subcategory generated by one object is semisimple by the decomposition theorem; its Tannaka group is therefore reductive. We denote by δ_0 the skyscraper perverse sheaf at the origin; its image is the tensor unit for convolution.

For an integral subvariety $Z \subset A$ of dimension d, let

$$\mathrm{IC}_Z = j_{!*}\mathbf{Q}_{Z_{\mathrm{reg}}}[d], \qquad j : Z_{\mathrm{reg}} \hookrightarrow A,$$

be its perverse intersection complex, extended by zero to A. If $\chi(\mathrm{IC}_Z) > 0$, we denote by $G(\mathrm{IC}_Z)$ the Tannaka group of the category generated by $\mathrm{IC}_Z \otimes_{\mathbf{Q}} \mathbf{C}$ in the displayed quotient, and by $\omega(\mathrm{IC}_Z)$ its tautological module. Then

$$\dim \omega(\mathrm{IC}_Z) = \chi(\mathrm{IC}_Z).$$

The action on the tautological module is faithful by construction. We suppress the scalar extension from notation inside Tannakian expressions. We write IC^H_Z for the polarizable pure Hodge module whose underlying perverse $\mathbf{Q}$-sheaf is IC_Z.

The commutativity constraint contains the usual cohomological sign. Since the support in this paper has even dimension two, the alternating projector on $\mathrm{IC}_X * \mathrm{IC}_X$ is carried by the fibre functor to the ordinary exterior square of $\omega(\mathrm{IC}_X)$. We write $\mathrm{Alt}^2(\mathrm{IC}_X)$ for its image in the Tannakian quotient.

3. The cyclic theta section

Until Section 5, fix $b \in \mathcal{B}(\mathbf{C})$ and suppress it from the notation. We write

$$E : y^2 = f_3(x), \qquad C : w^6 = f_3(x), \qquad y = w^3,$$

where f_3 is a separable cubic, and $\pi : C \to E$ is the resulting cyclic triple cover.

3.1. **The cover, the Prym, and the polarization.** The three branch points of π are the nonzero points of $E[2]$. If $\eta = \mathcal{O}_E(O)$, then the cyclic covering algebra is

$$\pi_*\mathcal{O}_C = \mathcal{O}_E \oplus \eta^{-1} \oplus \eta^{-2}.$$

Riemann–Hurwitz gives $g(C) = 4$, while the cyclic-cover canonical bundle formula gives

$$K_C = \pi^*(K_E \otimes \eta^2) = \pi^*\eta^2 = \kappa^2, \qquad \kappa = \pi^*\eta.$$

Projection formula yields

$$h^0(C, \kappa) = h^0(E, \eta) + h^0(E, \mathcal{O}_E) + h^0(E, \eta^{-1}) = 2.$$

The pullback $\pi^* : J(E) \to J(C)$ is injective. Indeed, if $\alpha \in \mathrm{Pic}^0(E)$ and $\pi^*\alpha \simeq \mathcal{O}_C$, then

$$1 = h^0(C, \pi^*\alpha) = h^0(E, \alpha) + h^0(E, \alpha\eta^{-1}) + h^0(E, \alpha\eta^{-2}),$$

and the last two terms vanish; hence $\alpha = \mathcal{O}_E$. Duality now shows that $\ker(\mathrm{Nm}_\pi)$ is connected. We therefore write

$$P = \ker(\mathrm{Nm}_\pi).$$

It has dimension three. If τ generates the deck group, then $\pi^* \mathrm{Nm}_\pi = 1 + \tau + \tau^2$ on $J(C)$. Hence $(1+\tau+\tau^2)(1-\tau) = 0$, and the injectivity of π^* gives $\mathrm{Im}(1-\tau) \subset P$. The invariant part of $H^0(C, \Omega^1_C)$ is $\pi^* H^0(E, \Omega^1_E)$ and has dimension one; hence $\mathrm{Im}(1-\tau)$ has dimension three. It is connected, so

$$P = \mathrm{Im}(1-\tau) = \ker(\mathrm{Nm}_\pi). \tag{3.1}$$

This identifies our convention with the Prym used in the polarization formula below.

Use the theta characteristic κ to centre the theta divisor on $J(C) = \mathrm{Pic}^0(C)$:

$$\Theta_\kappa = \{M \in J(C) : h^0(C, \kappa \otimes M) > 0\}.$$

Let $L = \mathcal{O}_{J(C)}(\Theta_\kappa)|_P$. Since $\mathrm{div}_E(y) = b_1 + b_2 + b_3 - 3O$, the branch divisor is reduced and belongs to $|\eta^3|$. The standard polarization formula for a connected degree-three cyclic cover with this branch data gives

$$\mathrm{type}(L) = (1,1,3), \qquad L^3 = 3!\,\chi(L) = 18;$$

see [16, Section 4] and [2, Corollary 12.1.4 and Lemma 12.3.1]. Scheme-theoretically,

$$X = \Theta_\kappa \cap P = \{M \in P : h^0(C, \kappa \otimes M) > 0\}.$$

As a determinantal divisor, the theta divisor is preserved as a closed subscheme by the Serre-duality involution $L' \mapsto K_C \otimes (L')^{-1}$ on $\mathrm{Pic}^3(C)$. Since $\kappa^2 = K_C$, its expression in the above centred coordinates is $M \mapsto M^{-1}$. Pulling back to P therefore gives, as closed subschemes,

$$X = -X. \tag{3.2}$$

This symmetry is centred at the origin. The uniqueness of this centre will follow from (3.7).

3.2. **The norm fibre and the resolution.** Let

$$n : C^{(3)} \longrightarrow \mathrm{Pic}^3(E), \qquad D \longmapsto \mathcal{O}_E(\pi_* D),$$

and put

$$N = n^{-1}(\eta^3), \qquad a : N \longrightarrow X, \quad D \longmapsto \mathcal{O}_C(D) \otimes \kappa^{-1}.$$

The map a is surjective on closed points. Indeed, if $M \in X$, a nonzero section of $\kappa \otimes M$ has an effective divisor D of degree three, and

$$\mathrm{Nm}_\pi \mathcal{O}_C(D) = \mathrm{Nm}_\pi(\kappa \otimes M) = \eta^3,$$

because $M \in P$ and $\mathrm{Nm}_\pi(\pi^*\eta) = \eta^3$. Hence $D \in N$ and $a(D) = M$. The finite morphism π induces a finite morphism $C^{(3)} \to E^{(3)}$; its base change over $|\eta^3| \subset E^{(3)}$ is the finite surjection $N \to |\eta^3| \simeq \mathbf{P}^2$. In particular $\dim X \le \dim N = 2 < 3 = \dim P$, and therefore $X \neq P$. Thus the restriction of the centred theta section to P is nonzero; consequently $X \in |L|$ is an effective Cartier divisor and is ample.

Lemma 3.1 (The norm fibre)**.** *Let $\Gamma = |\kappa| \simeq \mathbf{P}^1$ and $D_0 = r_1 + r_2 + r_3 \in \Gamma$, where r_i are the ramification points of π; put $b_i = \pi(r_i)$. The surface N has the unique singular point D_0, and the germ (N, D_0) is a simple elliptic singularity of type $\widetilde{E}_6$. If $\beta : \widehat{N} \to N$ is the blowup of D_0, then $\widehat{N}$ is smooth, its exceptional curve E_0 is elliptic with $E_0^2 = -3$, and the strict transform $\widetilde{\Gamma}$ is a (-1)-curve meeting*

E_0 transversely in one point. Contracting $\widetilde{\Gamma}$ gives a smooth surface S and a proper morphism $\rho : S \to X$. The image E_{exc} of E_0 has self-intersection -2 and maps to the origin of X.

Proof. In local coordinates at the three branch points, the norm map is analytically

$$(x_1, x_2, x_3) \longmapsto x_1^3 + x_2^3 + x_3^3.$$

Thus (N, D_0) is the affine cone over a smooth plane cubic, a simple elliptic singularity of type $\widetilde{E}_6$ in the standard classification [17, 20].

We next check that there are no other singularities. Away from the ramification points the differential of the norm map is nonzero. At a ramification point, the local covering is $z \mapsto z^3$. For a divisor with two points colliding there, in elementary symmetric coordinates one has

$$z_1^3 + z_2^3 = e_1^3 - 3e_1e_2,$$

which has no linear term; with three points colliding, however,

$$z_1^3 + z_2^3 + z_3^3 = e_1^3 - 3e_1e_2 + 3e_3$$

has a nonzero linear term. A degree-three divisor containing any unramified point has a nonzero differential in that point's direction. If all support points are ramification points, the multiplicity partitions are $1 + 1 + 1$, $2 + 1$, and 3; the last is noncritical by the $3e_3$ term above. Thus the only critical candidates in N are D_0 and $2r_i + r_j$ with $i \neq j$. The latter would require $2b_i + b_j \sim 3O$. Since the b_i are the nonzero two-torsion points, this would give $b_j = O$, a contradiction. Thus D_0 is the unique singular point of N.

Let $\beta : \widehat{N} \to N$ be the blowup of the vertex. Then $\widehat{N}$ is smooth, with exceptional elliptic curve E_0 satisfying $E_0^2 = -3$. The strict transform $\widetilde{\Gamma}$ meets E_0 transversely once. Indeed, locally along the pencil Γ, a moving member is $\operatorname{div}(w - t)$. At $t = 0$ the three local coordinates at the ramification points depend linearly and nontrivially on t. Thus Γ has multiplicity one at D_0, and its strict transform meets the exceptional cubic in one reduced point.

Let $a_C : C^{(3)} \to J(C)$ be the Abel–Jacobi map and let Θ_C be the principal theta divisor. The symmetric-product canonical formula [19] gives $K_{C^{(3)}} \equiv a_C^*\Theta_C$. Since N is a fibre of $n : C^{(3)} \to \mathrm{Pic}^3(E)$ and the target is a smooth curve, its normal line bundle is trivial. Adjunction and restriction to the Prym translate therefore give $K_N \equiv a^*L$. Since a contracts Γ, one has $K_N \cdot \Gamma = 0$. The cone has multiplicity three, and its discrepancy is -1; hence

$$K_{\widehat{N}} = \beta^*K_N - E_0.$$

Since $E_0 \cdot \widetilde{\Gamma} = 1$, this gives $K_{\widehat{N}} \cdot \widetilde{\Gamma} = -1$. Adjunction on $\widetilde{\Gamma} \simeq \mathbf{P}^1$ reads

$$-2 = (K_{\widehat{N}} + \widetilde{\Gamma}) \cdot \widetilde{\Gamma} = -1 + \widetilde{\Gamma}^2,$$

so $\widetilde{\Gamma}^2 = -1$. Contracting $\widetilde{\Gamma}$ produces a smooth surface S; the image E_{exc} of E_0 satisfies

$$E_{\mathrm{exc}}^2 = E_0^2 + (E_0 \cdot \widetilde{\Gamma})^2 = -3 + 1 = -2.$$

The composite $\widehat{N} \to N \to X$ is constant on $\widetilde{\Gamma}$, so it factors through a proper morphism $\rho : S \to X$; the curve E_{exc} maps to the origin. We identify this morphism as the minimal resolution below. $\square$

Lemma 3.2 (Euler stratification)**.** *The surfaces in Lemma 3.1 satisfy*

$$e(N) = 10, \qquad e(\widehat{N}) = 9, \qquad e(S) = 8.$$

Proof. We compute $e(N)$ from the finite map $N \to |\eta^3| \simeq \mathbf{P}^2$. Stratifying a line section of the plane cubic by its multiplicity pattern gives the following table. Here $B_{\mathrm{br}} = \{b_1, b_2, b_3\}$ is the branch divisor. Let $\mathfrak{D} \subset |3O|$ be the dual sextic parametrizing tangent lines. Its normalization is the Gauss map $E \to \mathfrak{D}$. A smooth plane cubic has no bitangent line by Bézout, so this map is bijective and $e(\mathfrak{D}) = e(E) = 0$.

For each i, let $L_i \simeq \mathbf{P}^1$ be the pencil of lines through b_i. Set-theoretically, $L_i \cap \mathfrak{D}$ consists of the tangent at b_i and the four tangents at the solutions of $2q = -b_i$. These five points are distinct. Hence $e(L_i \setminus \mathfrak{D}) = -3$. The three pencils have the branch line as their common point, and therefore

$$e\left(\bigcup_i (L_i \setminus \mathfrak{D})\right) = 3(-3) - 3 + 1 = -11.$$

Removing the branch line gives Euler characteristic -12 for transverse sections containing exactly one branch point. Since $e(|3O| \setminus \mathfrak{D}) = 3$, the transverse stratum avoiding the branch divisor has Euler characteristic $3 - (-11) = 14$.

Under the Gauss normalization, the tangent strata are the nine flexes, the three points $q = b_i$, the twelve solutions of $2q = -b_i$, and their complement. Their Euler characteristics are $9, 3, 12, -24$, respectively. The sets are disjoint: the flexes are $E[3]$, the b_i are nonzero points of $E[2]$, and multiplication by two is étale of degree four. Thus the displayed tangent lines are all distinct.

base divisor type	e(stratum)	# fibre	contribution
three distinct points, no branch point	14	27	378
three distinct points, one branch point	-12	9	-108
the branch divisor	1	1	1
$2q + r,\ q, r \notin B_{\mathrm{br}}$	-24	18	-432
$2b_i + r$	3	3	9
$2q + b_i$	12	6	72
$3q$ at a flex	9	10	90
			10

For an unramified point occurring with multiplicity m, the number of effective lifts is $\binom{m+2}{2}$; at a branch point there is only one lift. This gives, in the order of the table, the fibre cardinalities $27, 9, 1, 18, 3, 6, 10$. Blowing up D_0 replaces a point by the elliptic curve E_0, while contracting $\widetilde{\Gamma} \simeq \mathbf{P}^1$ replaces that curve by a point. Together with the critical-point calculation above, this proves

$$\begin{aligned} & e(N) = 10, \\ & e(\widehat{N}) = e(N) - 1 + e(E_0) = 9, \\ & e(S) = e(\widehat{N}) - e(\mathbf{P}^1) + 1 = 8. \end{aligned} \tag{3.3}$$

□

Proposition 3.3 (The theta surface and its resolution)**.** *The theta section $X \subset P$ is an integral normal symmetric ample Cartier divisor with* $\mathrm{Sing}(X) = \{0\}$. *The morphism $\rho : S \to X$ of Lemma 3.1 is its minimal resolution and is an isomorphism* *over $X \setminus \{0\}$. Its exceptional locus is the smooth elliptic curve E_{exc} with $E_{\mathrm{exc}}^2 = -2$; in particular, $(X, 0)$ is a simple elliptic singularity of type $\widetilde{E}_7$.*

Proof. The canonical map of C is birational: on the affine chart it is

$$(x, w) \longmapsto [w^2 : w : 1 : x].$$

Hence C is not hyperelliptic. Its canonical quadric is the rank-three quadric determined by the pencil $|\kappa|$, which has only one ruling. Indeed, by geometric Riemann–Roch, the divisors in any g_3^1 span lines in the canonical embedding, and their union is a ruling of the canonical quadric. Consequently κ is the unique g_3^1 on C. Riemann–Kempf [13] shows that the Abel map is an isomorphism over the $h^0 = 1$ locus. The corresponding points of N are smooth by the critical-point calculation above, so their images in X are smooth. The pencil $|\kappa|$ is the only positive-dimensional Abel fibre. It follows so far that $\mathrm{Sing}(X) \subset \{0\}$.

The Riemann singularity theorem [13] also gives $\mathrm{mult}_0(\Theta_\kappa) = h^0(C, \kappa) = 2$. Thus a local equation of Θ_κ at the origin has no linear term. Its restriction to the smooth threefold P is nonzero by the paragraph following the definition of $a : N \to X$, and it still has no linear term. The restricted divisor X is therefore singular at the origin. Consequently

$$\mathrm{Sing}(X) = \{0\}. \tag{3.4}$$

The smooth dense open $X \setminus \{0\}$ shows that the Cartier divisor X is generically reduced; since a Cartier divisor on the smooth variety P is Cohen–Macaulay, X is reduced. An ample divisor on an abelian variety is connected. If X were reducible, two components would meet in a curve, which would lie in $\mathrm{Sing}(X)$. Thus X is integral.

The reduced Cartier divisor X is Cohen–Macaulay and hence satisfies S_2; by (3.4) it is regular in codimension one. Serre's criterion therefore makes X normal. The uniqueness of the g_3^1 shows that the Abel map is an isomorphism away from Γ. Consequently $\rho : S \to X$ is a resolution whose only exceptional curve is E_{exc}. Since this curve is elliptic with self-intersection -2, the resolution is minimal, and the simple-elliptic classification identifies $(X, 0)$ as type $\widetilde{E}_7$ [17, 20]. □

Corollary 3.4 (The intersection complex)**.** *There is a decomposition*

$$R\rho_* \mathbf{Q}_S[2] \simeq \mathrm{IC}_X \oplus \delta_0.$$

Moreover, $\chi(\mathrm{IC}_X) = 7$ and $\mathrm{Stab}_P(X) = \{0\}$.

Proof. The map $\rho : S \to X$ is semismall, with the single relevant exceptional fibre E_{exc}. Its one-dimensional top intersection form is the nondegenerate matrix (-2). The semismall decomposition theorem [4, Theorem 3.4.1] gives

$$R\rho_* \mathbf{Q}_S[2] \simeq \mathrm{IC}_X \oplus \delta_0. \tag{3.5}$$

Taking Euler characteristics in (3.5) and using (3.3), we obtain

$$\chi(\mathrm{IC}_X) = 7. \tag{3.6}$$

The unique singular point also gives

$$\mathrm{Stab}_P(X) = \{0\}, \tag{3.7}$$

because a translation preserving X must preserve its singular locus. In particular, the origin is the unique centre of the symmetry (3.2): two distinct centres would compose to a nontrivial translation stabilizing X. □

Let $\Lambda_X \subset T^*P$ be the closure of the conormal bundle of X_{reg}, and let $\Lambda_0 = T_0^*P$ be the conormal fibre at the origin; both are regarded as conic Lagrangian cycles. Let

$$\gamma_X : \mathbf{P}(\Lambda_X) \longrightarrow \mathbf{P}(T_0^*P)$$

be the map induced by the cotangent projection.

Lemma 3.5 (Conormal degree)**.** *The characteristic cycle and conormal Gauss degree are*

$$CC(\mathrm{IC}_X) = \Lambda_X + \Lambda_0, \qquad \deg(\gamma_X) = 6.$$

In particular, the conormal Gauss map is dominant.

Proof. The normal form in the simple elliptic classification shows that a general hyperplane section of the $\widetilde{E}_7$ germ is of type A_3. Its sectional Milnor number is therefore 3, and the local Euler obstruction is $1-3=-2$ [17, 20, 3]. On the other hand, (3.5) gives $\chi((\mathrm{IC}_X)_0) = -1$. The Dubson–Kashiwara local index formula [12] therefore gives

$$CC(\mathrm{IC}_X) = \Lambda_X + m\Lambda_0, \qquad -1 = (-2) + m,$$

so $m = 1$. The Franecki–Kapranov index theorem [6, Theorem 1.3], together with (3.6), now gives

$$7 = \deg(\gamma_X) + \deg(\Lambda_0) = \deg(\gamma_X) + 1,$$

and hence

$$\deg(\gamma_X) = 6. \tag{3.8}$$

□

3.3. **The generic-twist Hodge vector.**

Lemma 3.6 (Surface invariants)**.** *The minimal resolution S satisfies*

$$K_S^2 = 16, \qquad e(S) = 8, \qquad \chi(\mathcal{O}_S) = 2.$$

Proof. Numerically, $K_N \equiv a^*(L|_X)$. Since $a : N \to X$ is birational and $X \in |L|$, the projection formula gives

$$K_N^2 = (a^*L)^2 = L^2 \cdot a_*[N] = L^2 \cdot [X] = L^3 = 18.$$

The blowup of the $\widetilde{E}_6$ vertex gives

$$K_{\widehat{N}}^2 = K_N^2 + E_0^2 = 15,$$

and $\widehat{N} \to S$ contracts a (-1)-curve, equivalently $\widehat{N}$ is the blowup of S at a smooth point. Hence $K_S^2 = K_{\widehat{N}}^2 + 1 = 16$. By Lemma 3.2, $e(S) = 8$, and Noether's formula gives $\chi(\mathcal{O}_S) = (K_S^2 + e(S))/12 = 2$. □

We next determine the Hodge cocharacter.

Put $U = P \setminus X$. Since X is ample, U is affine. Artin vanishing for the perverse sheaf $\mathbf{Q}_U[3]$ [1, Théorème 4.1.1] gives $H_c^i(U, \mathbf{Q}) = 0$ for $i < 3$. The long exact sequence for the pair (P, X) therefore identifies $H^1(P, \mathbf{Q})$ with $H^1(X, \mathbf{Q})$.

The surface X is normal, is a local complete intersection in P, and has only the isolated singular point 0. Hence $\mathbf{Q}_X[2]$ and IC_X are perverse, agree on X_{reg}, and their perverse kernel and cokernel are supported at 0. Point-supported perverse sheaves have hypercohomology only in degree zero, so

$$H^1(X, \mathbf{Q}) = \mathbb{H}^{-1}(X, \mathbf{Q}_X[2]) \xrightarrow{\sim} \mathbb{H}^{-1}(X, \mathrm{IC}_X) = IH^1(X, \mathbf{Q}).$$

Finally, taking degree -1 hypercohomology in (3.5) gives

$$H^1(P,\mathbf{Q}) \xrightarrow{\sim} IH^1(X,\mathbf{Q}) \xrightarrow{\sim} H^1(S,\mathbf{Q}).$$

Thus $q(S) = h^1(S,\mathcal{O}_S) = 3$ and $\mathrm{Pic}^0(P) \to \mathrm{Pic}^0(S)$ is an isogeny. The morphism $S \to P$ has a two-dimensional image, so S has maximal Albanese dimension.

Let $j : S \xrightarrow{\rho} X \hookrightarrow P$ be the composite morphism. For $\alpha \in \mathrm{Pic}^0(P)$, let $\mathcal{L}_\alpha$ be the unique unitary rank-one local system whose associated holomorphic line bundle is α. Define $\Omega \subset \mathrm{Pic}^0(P)$ by the conditions

$$H^i(S, K_S \otimes \alpha^{\pm 1}|_S) = 0 \quad (i > 0),$$
$$H^i(P, Rj_*\mathbf{Q}_S[2] \otimes \mathcal{L}_\alpha) = 0 \quad (i \neq 0).$$

Coherent generic vanishing gives a nonempty Zariski-open locus in $\mathrm{Pic}^0(P)$ for the first condition, while perverse generic vanishing gives a nonempty Zariski-open locus in the character variety for the second. Under the unitary correspondence, both restrict to Euclidean open dense subsets of the compact unitary character torus: for the second this uses the Zariski density of that torus, and for the first it uses the proper cohomology-support loci in $\mathrm{Pic}^0(P)$. Their intersection is therefore nonempty, so $\Omega \neq \varnothing$.

Proposition 3.7. *For every $\alpha \in \Omega$, the Hodge structure $H^0(P, \mathrm{IC}_X^H \otimes \mathcal{L}_\alpha)$ has Hodge numbers*

$$(h^{2,0}, h^{1,1}, h^{0,2}) = (2, 3, 2).$$

Equivalently, the associated antidiagonal Hodge cocharacter has weights $-2, 0, 2$ on $\omega(\mathrm{IC}_X)$ with multiplicities $2, 3, 2$, respectively.

Proof. Fix $\alpha \in \Omega$. Coherent generic vanishing [8] gives

$$H^i(S, K_S \otimes \alpha|_S) = 0 \qquad (i > 0).$$

Since $\chi(\mathcal{O}_S) = 2$, Riemann–Roch and Serre duality, applied also to α^{-1}, give

$$h^0(S, K_S \otimes \alpha|_S) = 2, \qquad h^2(S, \alpha|_S) = 2.$$

The complex $Rj_*\mathbf{Q}_S[2]$ is perverse: first $R\rho_*\mathbf{Q}_S[2]$ is perverse because ρ is semismall, and then extension along $X \hookrightarrow P$ is perverse t-exact. Generic vanishing for perverse sheaves on abelian varieties [15, Theorem 1.1] therefore gives

$$H^k(S, j^*\mathcal{L}_\alpha) = 0 \qquad (k \neq 2).$$

For a unitary local system, the Hodge decomposition is the twisted Dolbeault decomposition

$$H^{p,q}(S, j^*\mathcal{L}_\alpha) = H^q(S, \Omega_S^p \otimes \alpha|_S), \qquad p + q = 2.$$

The total dimension is $e(S) = 8$, and explicitly

$$h^{2,0} = h^0(S, K_S \otimes \alpha|_S) = 2, \qquad h^{0,2} = h^2(S, \alpha|_S) = 2, \qquad h^{1,1} = 8 - 2 - 2 = 4.$$

Thus the Hodge vector of $Rj_*\mathbf{Q}_S^H[2]$ is $(2, 4, 2)$. The Hodge-module refinement of (3.5), extended to P, is

$$Rj_*\mathbf{Q}_S^H[2] \simeq \mathrm{IC}_X^H \oplus \mathbf{Q}_0^H(-1).$$

The point summand has type $(1, 1)$ and removes one middle Tate class, leaving the Hodge vector $(2, 3, 2)$ for IC_X^H. $\square$

4. The generic pair-incidence splitting

We return to the family over $\mathcal{B}$. Let $\mathcal{E} \to \mathcal{B}$ be the Weierstrass elliptic scheme with zero section O, put $\eta = \mathcal{O}_{\mathcal{E}}(O)$, and let

$$\pi : \mathcal{C} \longrightarrow \mathcal{E}, \qquad \kappa = \pi^* \eta$$

be the relative cyclic cover and its distinguished theta characteristic. Let $\mathcal{P} \to \mathcal{B}$ be the relative Prym abelian scheme and let $\mathcal{X} \subset \mathcal{P}$ be the relative centred theta section. For any $Y \to \mathcal{B}$, undecorated symbols E_Y, C_Y, P_Y, X_Y, η and κ denote the corresponding base changes. The divisor $\mathcal{X}$ is flat over $\mathcal{B}$: it is a relative effective Cartier divisor in the smooth scheme $\mathcal{P}$, and its defining section is a non-zero-divisor on every fibre. For every closed point of $\mathcal{B}$, Proposition 3.3 identifies the fibre as integral with singular locus the origin. Geometric integrality is open in this flat proper family [9, IV_3, Théorème 12.2.4(viii)]; its complement has no closed points and is therefore empty. The fibrewise criterion for smoothness likewise shows that the relative nonsmooth locus is precisely the zero section. Set

$$\mathcal{M} = (\mathcal{X}/\mathcal{B})_{\mathrm{sm}} = \mathcal{X} \setminus 0.$$

The morphism $\mathcal{M} \to \mathcal{B}$ is smooth and surjective, and $\mathcal{M}$ is integral. Indeed, $\mathcal{M}$ is regular, so its irreducible components are open. Every component dominates the integral base and would give an irreducible component of the geometrically integral generic fibre. There can therefore be only one.

Let $\mathcal{N}$ be the relative norm fibre in $\mathcal{C}^{(3)}$. Every Abel class on $\mathcal{M}$ has $h^0 = 1$, and the relative Abel map identifies $\mathcal{N} \times_{\mathcal{X}} \mathcal{M}$ with $\mathcal{M}$. Thus the universal divisor on the relative symmetric cube gives a universal effective degree-three divisor over $\mathcal{M}$.

4.1. **A dominant normalized chart.**

Lemma 4.1 (Dominant normalized chart)**.** *There is an integral four-dimensional scheme $T_{\mathbf{Q}}$ over $\mathbf{Q}$ with rational parameters (m, e_1, e_2, e_3), where $m \neq 0$ and $\lambda = m^{-3}$. After base change to $\mathbf{C}$, the scheme*

$$T = T_{\mathbf{Q}} \times_{\mathbf{Q}} \mathbf{C}$$

is birational to $\mathcal{M}$, and suitable dense opens are isomorphic. On those opens the universal divisor D is given by (4.2). On $e_1 \neq 0$, put

$$\Lambda = e_1^3 \lambda, \qquad E_2 = \frac{e_2}{e_1^2}, \qquad E_3 = \frac{e_3}{e_1^3}, \qquad Z = e_1 z. \tag{4.1}$$

The residual-octic family and its factor-ordering cover are defined over $T_{\mathbf{Q}}$. Birationally, they are pulled back from their restrictions to the normalized chart $e_1 = 1$, with $e_1 \in \mathbf{G}_m$ as the extra parameter.

Proof. Let D be the unique reduced degree-three divisor representing a point in a dense open of $\mathcal{M}$. In the original short-Weierstrass coordinates, its norm is a line $y = mx + n$, with $m \neq 0$. Put

$$x' = mx + n, \qquad y' = y, \qquad w' = w.$$

The norm line is then $y' = x'$. If the three selected lifts have w'-coordinates a, b, c, then

$$g(x') - (x')^2 = \lambda(x' - a^3)(x' - b^3)(x' - c^3), \qquad \lambda = m^{-3}.$$

Put $e_i = e_i(a,b,c)$ and

$$p(W) = W^3 - e_1 W^2 + e_2 W - e_3.$$

Write

$$\Sigma_1 = e_1^3 - 3e_1e_2 + 3e_3, \qquad \Sigma_2 = e_2^3 - 3e_1e_2e_3 + 3e_3^2.$$

Conversely, the parameters reconstruct the curve and the divisor in the primed coordinates as

$$\begin{aligned} g(x') &= (x')^2 + \lambda\big((x')^3 - \Sigma_1(x')^2 + \Sigma_2 x' - e_3^3\big), \\ D: \quad x' &= y' = (w')^3, \qquad p(w') = 0. \end{aligned} \tag{4.2}$$

To make the map to the fixed short-Weierstrass family explicit, set

$$c_2 = 1 - \lambda\Sigma_1, \qquad c_1 = \lambda\Sigma_2, \qquad c_0 = -\lambda e_3^3, \qquad n = -\frac{c_2}{3\lambda}.$$

On the integral finite étale cubic cover defined by $m^3\lambda = 1$, the change $x' = mx+n$ gives

$$y^2 = x^3 + Ax + B, \qquad A = m\left(c_1 - \frac{c_2^2}{3\lambda}\right), \qquad B = g(n).$$

Conversely, a general point of $\mathcal{M}$ has a unique effective divisor D and its norm is cut out by a unique nonvertical line $y = mx + n$ on the open under consideration. The first coordinate change recovers m and the unordered triple of w'-coordinates recovers e_1, e_2, e_3. These two constructions are inverse. All displayed formulas are defined over $\mathbf{Q}$ and construct $T_{\mathbf{Q}}$. After base change to $\mathbf{C}$ and shrinking both domains, they identify T with a dense open of $\mathcal{M}$; in particular,

$$\mathbf{C}(T) = \mathbf{C}(\mathcal{M}). \tag{4.3}$$

The primed equations are equivariant for the scaling

$$(x', y', w') \longmapsto (r^3x', r^3y', rw'),$$

under which

$$(\lambda, e_1, e_2, e_3) \longmapsto (r^{-3}\lambda, re_1, r^2e_2, r^3e_3), \qquad m \longmapsto rm.$$

The corresponding bases of $H^0(C, 3\kappa - D)$ satisfy

$$s_0 \longmapsto r^3 s_0, \qquad s_1 \longmapsto r^3 s_1, \qquad s_2 \longmapsto r^4 s_2,$$

so the projective section coordinates transform by

$$[u : v : z] \longmapsto [u : v : r^{-1}z]. \tag{4.4}$$

Taking $r = e_1^{-1}$ gives precisely (4.1). Consequently, on the rational model,

$$\mathbf{Q}(\lambda, e_1, e_2, e_3) = \mathbf{Q}(\Lambda, E_2, E_3)(e_1),$$

and the residual equation, its rank-two locus, and its ordering character are pulled back under this purely transcendental extension, with section coordinate Z. Finally, if $q_0 = m/e_1$, then

$$\mathbf{Q}(T_{\mathbf{Q}}) = \mathbf{Q}(\Lambda, E_2, E_3)(q_0, e_1), \qquad q_0^3 = \Lambda^{-1}. \tag{4.5}$$

The same equality holds with $\mathbf{Q}$ replaced by $\mathbf{C}$ after base change. Thus the passage from the normalized model to $T_{\mathbf{Q}}$ adds the transcendental parameter e_1 and the displayed cubic extension. We henceforth drop the primes when referring to the normalized chart. $\square$

4.2. **The norm conic.** The formulas in this subsection are first computed on the rational model $T_{\mathbf{Q}}$; scalar extension to $\mathbf{C}$ is made only after the algebraic identities have been established. The evaluation map

$$H^0(C, 3\kappa) \longrightarrow H^0(D, (3\kappa)|_D)$$

has rank three. Since $h^0(C, 3\kappa) = 6$, its kernel $H^0(C, 3\kappa - D)$ has dimension three, with basis

$$s_0 = x - y, \qquad s_1 = p(w), \qquad s_2 = (x - e_3)w - e_1 y + e_2 w^2.$$

For $s = us_0 + vs_1 + zs_2$, write

$$s = \alpha + \beta w + \gamma w^2,$$

where

$$\begin{aligned}\alpha &= u(x - y) + v(y - e_3) - e_1 z y,\\ \beta &= e_2 v + z(x - e_3),\\ \gamma &= -e_1 v + e_2 z.\end{aligned}$$

Since $w^3 = y$, its cyclic norm is

$$\operatorname{Nm}(s) = \alpha^3 + \beta^3 y + \gamma^3 y^2 - 3\alpha\beta\gamma y. \tag{4.6}$$

The section (4.6) is divisible on E by the fixed line $y - x$. The quotient belongs to $H^0(E, 6O)$ and is represented by a conic

$$Q_s = q_{00} + q_{01}x + q_{02}y + q_{11}x^2 + q_{12}xy + q_{22}y^2.$$

Let M_s be its symmetric 3×3 matrix. Thus, with respect to the coordinates $(1, x, y)$, our convention is

$$M_s = \begin{pmatrix} q_{00} & q_{01}/2 & q_{02}/2 \\ q_{01}/2 & q_{11} & q_{12}/2 \\ q_{02}/2 & q_{12}/2 & q_{22} \end{pmatrix}.$$

Proposition 4.2 (Determinant factorization)**.** *Put $K_0 = \mathbf{Q}(\lambda, e_1, e_2, e_3)$. In the polynomial ring $K_0[u, v, z]$ one has*

$$\det(M_s) = (u - v)F_8, \tag{4.7}$$

where F_8 is homogeneous of degree eight and $u - v \nmid F_8$.

Proof. Substituting (4.6), reducing modulo $y^2 = g(x)$, and expressing the quotient in the standard basis of $H^0(E, 6O)$,

$$1,\ x,\ y,\ x^2,\ xy,\ y^2,$$

gives, by exact arithmetic in K_0, a homogeneous determinant of degree nine, divisible by $u - v$ with multiplicity one; the quotient is the asserted homogeneous octic. The reconstruction from the source equations is recorded in Section A. □

Lemma 4.3 (Geometric integrality of the residual octic)**.** *The projective curve*

$$\mathscr{R} = V(F_8) \subset \mathbf{P}^2_{K_0}$$

is geometrically integral.

Proof. Consider the specialization
$$(\lambda, e_1, e_2, e_3) = (1, 6, 11, 6), \qquad (a, b, c) = (1, 2, 3),$$
and define
$$H(u, v) = \frac{1}{8910} F_8(u, v, 1)\big|_{(\lambda, e_1, e_2, e_3) = (1,6,11,6)}.$$
The scalar is included to match the primitive integral specialization used below; it is nonzero modulo 13. Modulo 13, it has v-degree eight with constant leading coefficient -5, and its monic slice at $u = 1$ is
$$h(v) = v^8 + 3v^7 - 4v^6 - 3v^5 - 4v^4 - 5v^3 + 5v^2 - 2v - 1.$$
Exact polynomial arithmetic gives
$$\gcd\big(h, v^{13^4} - v\big) = 1, \qquad v^{13^8} - v \equiv 0 \pmod{h}.$$
Rabin's criterion shows that h is irreducible over $\mathbf{F}_{13}$. If $H = H_1 H_2$ with both factors nonconstant, the leading v-coefficients of $H_1, H_2 \in \mathbf{F}_{13}[u, v]$ would multiply to -5 and hence would both be units. Every factor of positive v-degree would retain that degree after $u = 1$, while a factor of v-degree zero would itself be a unit. Either case would contradict the irreducibility of h. Thus H is irreducible. Its total degree is eight, so homogenization introduces no component at infinity.

The same specialization contains the $\mathbf{F}_{13}$-point $(u, v) = (0, 2)$, where the gradient of H is $(9, 0)$. Equivalently, the primitive numerator $35640H$ used by the Singular wrapper has gradient $(-2, 0)$. If the irreducible projective octic were not geometrically irreducible, Frobenius would act transitively on its geometric components. The rational point would then lie on all components and would be singular, contrary to either displayed nonzero gradient. This special fibre is therefore geometrically integral.

To pass to characteristic zero, choose $R_0 = \mathbf{Z}[1/N]$ with $3 \mid N$ and $13 \nmid N$. There is an affine open
$$\mathscr{U} \subset \mathbb{A}^4_{R_0} = \operatorname{Spec} R_0[\lambda, e_1, e_2, e_3]$$
containing the displayed characteristic-13 parameter, on which every coefficient of F_8 is regular and one coefficient is invertible. Indeed, only finitely many denominator polynomials and one coefficient must be inverted, and the specialization just computed shows that they may be chosen nonzero at that point. After multiplying by a unit on $\mathscr{U}$, regard F_8 as a primitive homogeneous polynomial in $\Gamma(\mathscr{U}, \mathcal{O}_{\mathscr{U}})[u, v, z]$. It is then a fibrewise nonzero section of $\mathcal{O}_{\mathbf{P}^2_{\mathscr{U}}}(8)$. The sequence
$$0 \longrightarrow \mathcal{O}(-8) \xrightarrow{F_8} \mathcal{O} \longrightarrow \mathcal{O}_{V(F_8)} \longrightarrow 0$$
remains exact after base change; hence $V(F_8) \to \mathscr{U}$ is flat and projective. Geometric integrality is open in such a family [9, IV_3, Théorème 12.2.4(viii)]. Its open locus is nonempty by the special fibre just checked, and therefore contains the generic point of the integral scheme $\mathscr{U}$. The generic fibre is $\mathscr{R}$. □

Lemma 4.4 (Normalized and full function fields). *Work over $e_1 \neq 0$ and put*
$$K_{\mathrm{n}} = \mathbf{Q}(\Lambda, E_2, E_3),$$
with the normalized variables from (4.1). On the affine section chart $v \neq 0$, put
$$U = u/v, \qquad z_0 = z/v, \qquad \zeta = e_1 z_0.$$

Let $\mathscr{R}_{\mathrm{n}}$ be the residual curve obtained by setting $e_1 = v = 1$ and using coordinates $(\Lambda, E_2, E_3; U, \zeta)$. Let

$$\mathscr{R}_{T_{\mathbf{Q}}} = \mathscr{R} \times_{\operatorname{Spec} K_0} \operatorname{Spec} \mathbf{Q}(T_{\mathbf{Q}})$$

under the parameter map of Lemma 4.1. Then

$$k(\mathscr{R}) = k(\mathscr{R}_{\mathrm{n}})(e_1), \qquad k(\mathscr{R}_{T_{\mathbf{Q}}}) = k(\mathscr{R}_{\mathrm{n}})(e_1, q_0), \quad q_0^3 = \Lambda^{-1}. \tag{4.8}$$

The normalized curve $\mathscr{R}_{\mathrm{n}}$ is geometrically integral. After base change from $\mathbf{Q}$ to $\mathbf{C}$, the factor-ordering cover over the generic residual curve on T is the pullback of the corresponding cover over $\mathscr{R}_{\mathrm{n}}$ through the field extensions in (4.8).

Proof. In the original short-Weierstrass coordinates, the scaling used in Lemma 4.1 is induced by

$$(x, y, w) \longmapsto (r^2 x, r^3 y, rw), \qquad (A, B) \longmapsto (r^4 A, r^6 B).$$

In the primed coordinates it is $(x', y', w') \mapsto (r^3 x', r^3 y', rw')$. Equations (4.1) and (4.4) therefore identify the open set $e_1 v \neq 0$ in the full residual family birationally with $\mathbf{G}_m \times \mathscr{R}_{\mathrm{n}}$. This proves the first equality in (4.8). The chart $v \neq 0$ is dense: the geometrically integral degree-eight curve $\mathscr{R}$ cannot be the line $v = 0$.

By Lemma 4.3, the base change of $\mathscr{R}_{\mathrm{n}}$ to $K_{\mathrm{n}}(e_1)$ is geometrically integral. Geometric integrality descends through the faithfully flat extension $K_{\mathrm{n}} \subset K_{\mathrm{n}}(e_1)$, so $\mathscr{R}_{\mathrm{n}}$ is geometrically integral. The second equality in (4.8) follows from (4.5). Formation and ordering of the two factors of a rank-two conic commute with field extension, which proves the final assertion. □

Proposition 4.5 (Triviality of the ordering cover)**.** *On the normalized chart $e_1 = v = 1$, write $(\Lambda, E_2, E_3, U, \zeta)$ again as $(\lambda, e_2, e_3, u, z)$, put $K_1 = \mathbf{Q}(\lambda, e_2, e_3, u)$, and define*

$$\widetilde{F}(z) = \frac{4 \det(M_{us_0 + s_1 + zs_2})}{9e_2(e_2 - e_3)(u - 1)} \in K_1[z].$$

Let $\ell = \mathrm{lc}_z(\widetilde{F})$ and put $F = \ell^{-1}\widetilde{F}$. Then F is monic of degree six. If

$$\Delta = -4\left(q_{00}q_{11} - (q_{01}/2)^2\right),$$

there are $A, B \in K_1[z]$ with $\deg A = 3$, $\deg B \leq 2$, and

$$F = A^2 + 3\Delta B^2. \tag{4.9}$$

After extending scalars from $\mathbf{Q}$ to $\mathbf{C}$, the finite étale *double cover that orders the two distinct line factors of the generic rank-two conic on $\mathscr{R}$ is split.*

Proof. The exact coefficient formulas and two algebraically different checks of (4.9) are given in Section A. On the cover $t^2 + 3\Delta = 0$, the identity becomes

$$F = (A + tB)(A - tB).$$

The formula for B in Section A has nonzero z^2-coefficient. Since $\deg_z(B) \leq 2 < 6 = \deg_z(F)$ and F is irreducible by Lemma 4.4, B remains nonzero in the function field of $F = 0$. Likewise, A has degree three and is nonzero there. Thus $A/B \neq 0$, while (4.9) gives

$$-3\Delta = (A/B)^2 \neq 0.$$

The ordering algebra is therefore étale at the generic point, and its two rational roots $t = \pm A/B$ are distinct.

It remains to identify this double cover with the intrinsic ordering cover. The quantity Δ is the discriminant of the restriction of Q_s to a coordinate line. If a symmetric 3×3 matrix M has rank two, then $\operatorname{adj}(M) = caa^{\mathsf{t}}$ for a nonzero vector a and scalar c. Hence any two nonzero principal cofactors are ca_i^2 and ca_j^2; their ratio is a square. The square class obtained from any nonzero principal minor is consequently independent of the coordinate line and is the discriminant character that orders the two factors of the conic. Thus Δ defines the intrinsic ordering cover. The constant -3 is a square over $\mathbf{C}$, and Lemma 4.4 carries this splitting to the full generic residual curve and then to T. $\square$

4.3. **The scheme-theoretic Abel inverse.** Over the chart T of Lemma 4.1, let $p : C_T \to T$ be the relative curve and let D be its universal divisor. After restricting T, the sheaf $p_*\mathcal{O}_{C_T}(3\kappa - D)$ is locally free of rank three. The residual equation $F_8 = 0$ defines a relative octic

$$\mathscr{R}_T \subset \mathbf{P}_T\big(p_*\mathcal{O}_{C_T}(3\kappa - D)\big).$$

After a further shrinking of T, the section F_8 is nonzero on every projective-space fibre. Thus $\mathscr{R}_T \to T$ is a flat projective relative Cartier divisor. Its generic fibre is geometrically integral by Lemmas 4.3 and 4.4. Flatness forces every irreducible component to dominate the integral base T, so the integral generic fibre gives a single component. A nilpotent section would vanish generically and hence be $\mathcal{O}_T$-torsion, which flatness excludes. Thus $\mathscr{R}_T$ is integral.

Let $\mathscr{R}_T^{(2)} \subset \mathscr{R}_T$ be the open subset on which the residual conic has rank exactly two. The relative Hilbert scheme of line components of the universal conic is then finite étale of degree two over $\mathscr{R}_T^{(2)}$. Choosing one component determines the other and hence orders them; denote this cover by

$$\mathrm{Ord}_{\mathrm{fac}} \longrightarrow \mathscr{R}_T^{(2)}.$$

Its points will be written $([s], \ell_1, \ell_2)$, with $Q_s = \ell_1\ell_2$ up to a unit pulled back from the base.

Definition 4.6 (Admissible residual-norm locus)**.** Let $\mathscr{G} \subset \mathscr{R}_T^{(2)}$ be the open locus on which the following conditions hold after pullback to $\mathrm{Ord}_{\mathrm{fac}}$:

(i) the fixed divisor D is reduced;
(ii) the fixed norm line and the two lines $\ell_i = 0$ meet E transversely, their degree-three line sections are pairwise disjoint, and the residual line sections avoid the branch divisor of $C \to E$;
(iii) above each residual point, s vanishes on exactly one sheet of $C \to E$, and every such zero is simple;
(iv) the two resulting degree-three zero divisors have $h^0 = 1$, and their Abel classes lie in the relative smooth locus of $\mathcal{X}$.

These conditions define an open subset of $\mathrm{Ord}_{\mathrm{fac}}$ invariant under interchanging ℓ_1 and ℓ_2. Its image is open because the finite étale cover is open, and invariance makes it the full inverse image of that image. This defines $\mathscr{G}$.

Lemma 4.7 (Relative ordered-zero construction)**.** *Put*

$$\mathrm{Ord}_{\mathrm{fac},\mathscr{G}} = \mathrm{Ord}_{\mathrm{fac}} \times_{\mathscr{R}_T^{(2)}} \mathscr{G}.$$

Let $\mathrm{Ord}_{\mathrm{div}}$ *be the functor over* $\mathscr{G}$ *whose* W*-points are ordered pairs of relative effective Cartier divisors* (D_1, D_2) *of degree three on* C_W/W *such that*

$$\operatorname{div}(s_W) = D_W + D_1 + D_2$$

and the unordered pair $\{\pi_* D_1, \pi_* D_2\}$ *is the unordered pair of line sections of* E_W *defined by the two components of* Q_{s_W}*. This functor is represented by a finite étale double cover of* $\mathscr{G}$*. The assignment* $Q_s = \ell_1\ell_2 \mapsto (D_1, D_2)$ *is an isomorphism of finite étale double covers*

$$\mathrm{Ord}_{\mathrm{fac},\mathscr{G}} \xrightarrow{\sim} \mathrm{Ord}_{\mathrm{div}}\,.$$

It commutes with arbitrary base change, including nonreduced base change. Moreover, each D_i *is a relative effective Cartier divisor of degree three, and* $[\mathcal{O}(D_i)\otimes\kappa^{-1}]$ *lies in* $\mathcal{X}_{\mathrm{sm}/\mathcal{B}}$*.*

Proof. For the forward construction put $T' = \mathrm{Ord}_{\mathrm{fac},\mathscr{G}}$ and write $Q_s = \ell_1\ell_2$ in its universal ordering. By definition of $\mathscr{G}$, none of the tangency, branch, collision, multiple-zero, or Abel speciality loci excluded in Definition 4.6 occurs. For the remainder of the forward construction, every object with a subscript T is pulled back to T', and we suppress this base change from the notation. The line ℓ_i then cuts out a relative finite étale divisor $B_i \subset E_T$ of degree three. Moreover

$$U_i = C_T \times_{E_T} B_i \longrightarrow B_i$$

is finite étale of degree three, because B_i avoids the branch divisor.

We first select the zero over B_i scheme-theoretically. Étale-locally on B_i, split $U_i = B_i \sqcup B_i \sqcup B_i$ and trivialize the restriction of 3κ, using the induced trivialization of its norm line. If $a_1, a_2, a_3 \in \mathcal{O}_{B_i}$ are the three values of s, compatibility of the finite locally free norm with base change gives

$$a_1a_2a_3 = \mathrm{Nm}_\pi(s)|_{B_i} = 0 \tag{4.10}$$

as an equality in the structure sheaf, not merely at geometric points. The collision exclusions imply that locally two values, say a_2, a_3, are units. Equation (4.10) then forces $a_1 = 0$, also in nilpotent directions. Consequently

$$Z_i = Z(s) \times_{E_T} B_i \subset U_i$$

is locally exactly one open-and-closed sheet of U_i/B_i. Thus $Z_i \to B_i$ is finite étale of degree one and hence is an isomorphism.

Let t be a local equation for B_i in E_T. Since B_i occurs with multiplicity one in the norm divisor and the other norm factors are disjoint from it, $\mathrm{Nm}_\pi(s) = tu$ with u a unit along B_i. Étale-locally on a neighbourhood of B_i in the complement of the branch divisor, split the cover $C_T \to E_T$. On the selected sheet the other two values of s remain units, so the same local product formula shows that the selected value is t times a unit. The zero is therefore simple. The image $D_i \subset C_T$ of Z_i is finite étale of degree three over T'; as a finite étale multisection of the smooth relative curve C_T/T', it is a relative effective Cartier divisor.

The pairwise disjoint divisors D, D_1, D_2 lie in $Z(s)$ and have total relative degree nine, equal to $\deg(3\kappa)$. After dividing by their Cartier equations, the residual section has degree zero and is nonvanishing on every geometric fibre. Nakayama's lemma makes it a trivialization, so there is no vertical or nilpotent residual contribution and

$$Z(s) = D + D_1 + D_2 \tag{4.11}$$

as relative Cartier divisors. Conversely, an ordered divisor datum determines the unique projective section $[s]$ with zero divisor $D + D_1 + D_2$, and taking its norm recovers the ordered line sections B_i and hence the ordered factors ℓ_i. Thus these assignments are inverse already at the divisor level.

It remains to compare this divisor construction with the relative Abel incidence. Let

$$\alpha : C_T^{(3)} \longrightarrow \mathrm{Pic}^3(C_T/T)$$

be the relative Abel map and let $\mathcal{F} = \mathrm{Nm}^{-1}(\eta^3) \subset \mathrm{Pic}^3(C_T/T)$. The norm fibre is the cartesian base change

$$N_T = C_T^{(3)} \times_{\mathrm{Pic}^3(C_T/T)} \mathcal{F}.$$

Let $W_3^{(1)} \subset \mathrm{Pic}^3(C_T/T)$ be the open on which $h^0 = 1$ and the pushforward of a Poincaré bundle is locally free of rank one and commutes with base change. The standard projective-bundle description of the Abel map and cohomology and base change then identify its source with the projectivization of that rank-one bundle. Thus the map is an isomorphism and remains so after arbitrary base change. Fibrewise, this is the $h^0 = 1$ case of Riemann–Kempf [13]:

$$\alpha^{-1}(W_3^{(1)}) \xrightarrow{\sim} W_3^{(1)}.$$

Base changing this isomorphism by $\mathcal{F} \to \mathrm{Pic}^3(C_T/T)$ and translating by κ^{-1} gives

$$N_T^{(1)} \xrightarrow{\sim} X_T^{(1)}.$$

Here, by definition, $X_T^{(1)}$ is the open part of the theta section represented by line bundles with $h^0 = 1$. We replace both sides by the inverse images of $X_T^{(1)} \cap X_{T,\mathrm{sm}/T}$, as imposed above. Thus the divisor construction descends to the relative smooth Abel incidence.

Zero schemes, fibre products, finite locally free norms, and the cartesian Abel construction all commute with arbitrary base change. Hence both composites preserve the universal zero schemes and are the identity as morphisms. The use of $C_T^{(3)} \simeq \mathrm{Hilb}^3(C_T/T)$ is only representability; no normalization of the ordering cover or of pair fibres is involved. $\square$

Let $x_T : T \to \mathcal{M}$ be the point represented by the universal divisor D, so that $x_T = [\mathcal{O}(D) \otimes \kappa^{-1}]$. Define

$$\mathcal{I}_T^{\dagger} = \{y \in X_T^{(1)} \cap X_{T,\mathrm{sm}/T} : -x_T - y \in X_T^{(1)} \cap X_{T,\mathrm{sm}/T}\}.$$

The two relative Abel inverses give universal divisors D_1, D_2 on $C_T \times_T \mathcal{I}_T^{\dagger}$. Their Abel classes satisfy

$$[\mathcal{O}(D + D_1 + D_2) \otimes \kappa^{-3}] = 0 \quad \text{in } \mathrm{Pic}^0(C_T/T).$$

Thus $\mathcal{O}(D_1 + D_2)$ and $\mathcal{O}(3\kappa - D)$ differ only by a line bundle pulled back from the base. Their divisor sections determine a unique projective relative section

$$\vartheta : \mathcal{I}_T^{\dagger} \longrightarrow \mathbf{P}_T\big(p_*\mathcal{O}_{C_T}(3\kappa - D)\big).$$

Taking norms shows that ϑ factors through $\mathscr{R}_T$. Set

$$\mathcal{I}_T^{\mathrm{adm}} = \vartheta^{-1}(\mathscr{G}).$$

Proposition 4.8 (Abel-incidence identification)**.** *There is an isomorphism over* $\mathscr{G}$

$$\Phi : \mathrm{Ord}_{\mathrm{fac},\mathscr{G}} \xrightarrow{\sim} \mathcal{I}_T^{\mathrm{adm}}, \qquad ([s], \ell_1, \ell_2) \longmapsto [\mathcal{O}(D_1) \otimes \kappa^{-1}]. \tag{4.12}$$

Interchanging ℓ_1 and ℓ_2 corresponds to the involution $y \mapsto -x_T - y$. The isomorphism commutes with arbitrary base change.

Proof. On $\mathrm{Ord}_{\mathrm{fac},\mathscr{G}}$, put

$$y_i = [\mathcal{O}(D_i) \otimes \kappa^{-1}] \in P_T.$$

The divisor equality (4.11) gives

$$x_T + y_1 + y_2 = [\mathcal{O}(D + D_1 + D_2) \otimes \kappa^{-3}] = 0. \tag{4.13}$$

Hence $y_2 = -x_T - y_1$, and the ordered-zero construction of Lemma 4.7 defines the morphism Φ. The defining conditions of $\mathscr{G}$ show that it lands in $\mathcal{I}_T^{\mathrm{adm}}$.

Conversely, let y be a W-point of $\mathcal{I}_T^{\mathrm{adm}}$, write $p_W : C_T \times_T W \to W$ for the projection, put $y_1 = y$ and $y_2 = -x_T - y$, and let D_1, D_2 be their relative Abel inverse divisors. Reading (4.13) from right to left gives

$$\mathcal{O}(D_1 + D_2) \simeq \mathcal{O}(3\kappa - D) \otimes p_W^* \mathcal{N}$$

for a line bundle $\mathcal{N}$ on W. The divisor section therefore gives a unique projective section $[s]$ with $\mathrm{div}(s) = D + D_1 + D_2$; the factor pulled back from W does not affect this projective point. Since $y_i \in P_T$,

$$\mathrm{Nm}_\pi \mathcal{O}(D_i) = \mathrm{Nm}_\pi(\kappa) = \eta^3.$$

Thus the effective divisor $\pi_* D_i$ on E_T is cut out by a unique projective line ℓ_i. Multiplicativity of the finite locally free norm gives, up to a unit from W,

$$\mathrm{Nm}_\pi(s) = \ell_D \ell_1 \ell_2, \qquad Q_s = \ell_1 \ell_2,$$

where ℓ_D is the fixed norm line of D. Admissibility makes the two residual factors distinct and orders them by the ordered pair (D_1, D_2). This constructs the inverse of (4.12).

The relative Abel inverse on the $h^0 = 1$ locus, formation of zero divisors, and finite locally free norms commute with arbitrary base change. The two composites preserve the universal Cartier divisors and hence are the identity as morphisms. Swapping D_1, D_2 gives $y_1 \leftrightarrow y_2$, which is the stated incidence involution. □

Lemma 4.9 (Nonemptiness of the admissible locus)**.** *The open subscheme $\mathscr{G}$ of Definition 4.6 is nonempty in characteristic zero. It is dense in $\mathscr{R}_T$, dominates T, and its geometric generic fibre is a nonempty dense open of the geometric generic residual curve.*

Proof. After reduction modulo 13, an exact witness in the displayed universal basis (s_0, s_1, s_2) is provided by

$$E : y^2 = x^3 - 64x + 64, \quad (a, b, c) = (1, 2, -2), \quad (u : v : z) = (10 : 9 : 0) \pmod{13}.$$

At this point the conic has rank two and factors over $\mathbf{F}_{13}$ as

$$Q_s = 4 + 3x + y + 2x^2 + 3xy + 12y^2 = -(4x + y - 6)(6x + y + 5).$$

The sample script uses an auxiliary kernel basis. If (U, V, Z) are its coordinates, then

$$(u, v, z) = (U + 5V + Z,\ 4V + Z,\ -V).$$

Thus its internal point $(U : V : Z) = (1 : 0 : 9)$ is exactly the displayed universal point $(u : v : z) = (10 : 9 : 0)$; the verifier checks this change of basis before recording the values below. The two line-section discriminants, the branch resultants, the pair resultant, and the six evaluations on the fixed divisor are all nonzero; see

Section A. The canonical-model calculation used in Proposition 3.3 remains valid modulo 13: the canonical map is

$$(x,w) \longmapsto [w^2 : w : 1 : x],$$

and its unique canonical quadric is the rank-three cone $X_0X_2 - X_1^2 = 0$. Its unique ruling therefore makes $|\kappa|$ the unique g^1_3. In these coordinates, the norm of a section of $H^0(C,\kappa) = H^0(E,\eta) \oplus H^0(E,\mathcal{O}_E)$ cuts out a line with no x-term, whereas the three norm lines

$$y - x, \qquad 4x + y - 6, \qquad 6x + y + 5$$

all have a nonzero x-term. Thus their divisors have $h^0 = 1$ and avoid the contracted pencil. The local norm-fibre differential computation in Lemma 3.1 is algebraic and remains valid in characteristic 13, where 2 and 3 are invertible. On the $h^0 = 1$ locus the Abel map is the projectivization of a rank-one bundle and hence is an isomorphism, as in Lemma 4.7. The displayed divisors consequently belong to the relative smooth theta locus, as required in Definition 4.6(iv). At every residual point the branch avoidance makes $C \to E$ étale, while transversality makes the vanishing order of the norm equal to one. The three nonnegative vanishing orders of s on the three sheets sum to one. Hence exactly one sheet contains a simple zero, which verifies Definition 4.6(iii).

We now justify the passage from this point to characteristic zero. Enlarge the localization $R_0 = \mathbf{Z}[1/N]$ used in Lemma 4.3, without inverting 13, so that the chart $T_{\mathbf{Q}}$ and all the conditions in Definition 4.6 spread out. Because $v = 9$ at the witness, work on the affine chart $v \neq 0$. Let A_0 be the coordinate ring of an affine open in the resulting R_0-model of $T_{\mathbf{Q}}$ that contains the witness. It is a localization of

$$R_0[m, m^{-1}, e_1, e_2, e_3]$$

and is therefore a unique factorization domain. Put

$$U_0 = u/v, \qquad Z_0 = z/v.$$

Clear denominators from $F_8(U_0, 1, Z_0)$ and divide by the content to obtain a primitive polynomial $f \in A_0[U_0, Z_0]$.

The polynomial f is irreducible over $\mathrm{Frac}(A_0)$: its projective zero locus is the base change of the geometrically integral curve in Lemmas 4.3 and 4.4. Gauss's lemma therefore makes f irreducible in $A_0[U_0, Z_0]$. Hence

$$\mathscr{Y} = \operatorname{Spec} A_0[U_0, Z_0]/(f)$$

is integral. It is also R_0-flat, because its coordinate ring is R_0-torsion-free and R_0 is a localization of $\mathbf{Z}$.

The admissibility conditions define an open subscheme of $\mathscr{Y}$, and the displayed characteristic-13 point belongs to it. Since $\mathscr{Y}$ is affine and noetherian, there is a principal open $D(g)$ containing that point and contained in the admissible locus. In particular, g is nonzero in the domain $\Gamma(\mathscr{Y}, \mathcal{O}_{\mathscr{Y}})$, so $D(g)$ contains the generic point of $\mathscr{Y}$. Its characteristic-zero fibre is therefore nonempty and dominates the generic point of $T_{\mathbf{Q}}$. After base change to $\mathbf{C}$, it is contained in $\mathscr{G}$. Thus $\mathscr{G}$ is nonempty, dense in the integral scheme $\mathscr{R}_T$, and dominates T. Its geometric generic fibre is a nonempty open of a geometrically integral curve and hence is dense. $\square$

Corollary 4.10 (The two admissible branches)**.** *After replacing $\mathscr{G}$ by a dense open subset, there is an isomorphism over $\mathscr{G}$*

$$\mathcal{I}_T^{\mathrm{adm}} \simeq \mathscr{G} \sqcup \mathscr{G}.$$

The incidence involution exchanges the two copies. On a geometric generic fibre over T, each copy is a dense open of a geometrically integral curve.

Proof. By Proposition 4.5 and Lemma 4.4, the finite étale factor-ordering cover has two rational sections at the generic point of $\mathscr{G}$, given on the principal-minor chart by $t = \pm A/B$. They extend to two disjoint sections after shrinking $\mathscr{G}$, so $\mathrm{Ord}_{\mathrm{fac},\mathscr{G}} \simeq \mathscr{G} \sqcup \mathscr{G}$. Apply Proposition 4.8. The final assertion follows from Lemmas 4.3 and 4.9. □

4.4. **Boundary exhaustion and global labels.** Let ξ be the generic point of the integral scheme $\mathcal{M}$, and let $\bar{\xi}$ be a geometric generic point. For a geometric point x of $\mathcal{M}$, define the pair fibre scheme-theoretically by

$$C_x = X \cap (-x - X) \subset P, \qquad C_x^{\mathrm{ne}} = C_x \setminus \{0, -x\}.$$

Over $\bar{\xi}$, the unique effective divisors representing $x, y, -x-y$ satisfy

$$D_x + D_y + D_{-x-y} = \mathrm{div}(s_y) \tag{4.14}$$

for a projective section $[s_y] \in \mathbf{P}H^0(C, 3\kappa - D_x)$. This defines a rational map

$$\Psi : C_x \dashrightarrow \mathbf{P}H^0(C, 3\kappa - D_x).$$

Lemma 4.11 (Boundary exhaustion)**.** *The rational map Ψ has finite fibres wherever it is defined. The determinant component $u = v$ gives no nonexceptional incidence-curve component; its generic divisor data give only the Abel classes $y = 0, -x$. Every irreducible curve component of $C_{\bar{\xi}}^{\mathrm{ne}}$ dominates the residual octic $\mathscr{R}$, meets its admissible open, and is the closure of one of the two branches in Corollary 4.10. In particular, $C_{\bar{\xi}}^{\mathrm{ne}}$ has exactly two irreducible components.*

Proof. For a fixed section, the residual zero-cycle has only finitely many degree-three subcycles, so Ψ has finite fibres. In the basis used for (4.7), put

$$h = x - e_1 w^2 + e_2 w - e_3.$$

Since $y = w^3$ on C, direct substitution gives

$$s_0 + s_1 = h, \qquad s_2 = wh. \tag{4.15}$$

The zero divisor of $h \in H^0(C, 2\kappa)$ is $D_x + D_{-x}$. Indeed, it contains D_x, and the residual degree-three divisor has Abel class $-x$ and is unique on the generic $h^0 = 1$ locus. Equation (4.15) identifies the line $u = v$ with

$$\mathbf{P}\big(h \cdot H^0(C, \kappa)\big) = \mathbf{P}H^0(C, 3\kappa - D_x - D_{-x}).$$

Its members have divisor $D_x + D_{-x} + D_\kappa$ with $D_\kappa \in |\kappa|$. For a general member of this pencil, $\pi_* D_{-x}$ and $\pi_* D_\kappa$ are reduced, disjoint line sections of the plane cubic E. A degree-three subcycle using two points from the first section and one from the second cannot itself be a line section: a line through the first two points is the first line and cannot contain the third point. The same argument applies with the roles reversed. Hence the only degree-three subcycles whose pushforward is a line section are the two complete triples. Their Abel classes are $-x$ and 0.

The exceptional parameters of the structural pencil form a finite set. If an incidence-curve component mapped dominantly to $u = v$, its generic point would contradict the preceding paragraph; if it mapped to one parameter, it would contradict the finiteness of the fibres of Ψ. Thus $u = v$ gives no nonexceptional curve component.

The same two-line argument shows that, over $\mathscr{G}$ the only Prym degree-three subcycles of the six residual zeros are the two complete line sections. By Proposition 4.8 and Corollary 4.10, these give exactly the two displayed admissible branches.

The indeterminacy of Ψ is contained in the two exceptional points. By (4.7), every nonexceptional curve component therefore maps to the residual octic. Its image is one-dimensional because Ψ has finite fibres, and hence is dense because $\mathscr{R}$ is geometrically integral. It must meet the dense open $\mathscr{G}$. An irreducible curve cannot contain both disjoint admissible branches as open subsets, so their closures are exactly the two nonexceptional components. Here we use $\mathbf{C}(T) = \mathbf{C}(\mathcal{M})$ from (4.3) to identify the two geometric generic fibres. $\square$

Proposition 4.12 (The geometric generic pair fibre)**.** *The scheme $C_{\bar{\xi}}$ is reduced, and $C_{\bar{\xi}}^{\mathrm{ne}}$ has exactly two geometrically integral irreducible components $A_{\bar{\xi}}$ and $B_{\bar{\xi}}$. They are exchanged by $y \mapsto -x - y$, and both descend to $k(\xi)$.*

Proof. By Lemmas 4.3, 4.9 and 4.11, Propositions 4.5 and 4.8, and Corollary 4.10, the nonexceptional geometric generic fibre has exactly two geometrically integral components, exchanged by the incidence involution. Since $\mathrm{Stab}_P(X) = \{0\}$ and x is generic, the integral Cartier divisors X and $-x - X$ are distinct. Their local equations form a regular sequence in the smooth threefold P, so $C_{\bar{\xi}}$ is a pure local complete intersection curve. It is therefore Cohen–Macaulay and has no embedded associated points. The two dense admissible branches show that it is reduced at every generic point. Hence $C_{\bar{\xi}}$ is reduced.

It remains to descend the two labels. Let τ be the order-three deck transformation. On P one has

$$1 + \tau + \tau^2 = 0,$$

and $\tau(X) = X$. Therefore

$$y_+(x) = \tau x \in C_x, \qquad s_+(x) = (x, \tau x)$$

defines a rational section of the pair-incidence family; the associated ordered pair is $(\tau x, \tau^2 x)$. This section meets the smooth locus of the projection of the pair-incidence family to the x-coordinate on a nonempty open. Otherwise the Gauss covectors at τx and $\tau^2 x$ would agree for general x. Equivariance would then put the general Gauss image in the projective fixed locus of τ, contrary to (3.8). Explicitly,

$$T_0^* P = \left\langle \frac{dx}{w^4} \right\rangle \oplus \left\langle \frac{dx}{w^5}, \frac{x\,dx}{w^5} \right\rangle,$$

and the two summands have distinct τ-eigenvalues. The projective fixed locus is a point union a line, a proper subset of $\mathbf{P}^2$.

Over $k(\xi)$, the generic value of y_+ is a smooth rational point and lies on exactly one geometric component. Galois therefore fixes the component containing y_+ and, since there are exactly two components, also fixes the other. This proves descent. $\square$

5. Spreading the generic splitting

Define the relative pair-incidence scheme

$$\mathcal{I} = \{(x, y) \in \mathcal{M} \times_{\mathcal{B}} \mathcal{X} : -x - y \in \mathcal{X}\},$$

with the scheme structure induced by the two theta equations, and let $q : \mathcal{I} \to \mathcal{M}$ be the first projection. The equations $y = 0$ and $y = -x$ define two sections of q. We write $\mathcal{I}^{\mathrm{ne}}$ for their complement.

Theorem 5.1 (Uniform pair splitting)**.** *There are a nonempty open subscheme $\mathcal{M}^\circ \subset \mathcal{M}$ and closed subschemes $\mathcal{A}_1, \mathcal{A}_2 \subset \mathcal{I}|_{\mathcal{M}^\circ}$ with the following properties.*

- (i) *Each $\mathcal{A}_i \to \mathcal{M}^\circ$ is flat, and every geometric fibre is a geometrically integral curve.*
- (ii) *Every geometric fibre of $q|_{\mathcal{M}^\circ}$ is reduced, and*
$$\mathcal{I}^{\mathrm{ne}}|_{\mathcal{M}^\circ} = (\mathcal{A}_1 \cup \mathcal{A}_2) \cap \mathcal{I}^{\mathrm{ne}}$$
scheme-theoretically.
- (iii) *The two fibre curves are distinct at every geometric point of $\mathcal{M}^\circ$, and the involution $(x, y) \mapsto (x, -x - y)$ exchanges $\mathcal{A}_1$ and $\mathcal{A}_2$.*

The image

$$\mathcal{B}^\circ = \mathrm{im}(\mathcal{M}^\circ \to \mathcal{B})$$

is a nonempty Zariski-open subset. For every $b \in \mathcal{B}^\circ(\mathbf{C})$, the fibre

$$U_b = \mathcal{M}^\circ \times_{\mathcal{B}} \{b\}$$

is a nonempty dense connected open subset of $(X_b)_{\mathrm{reg}}$. For every $x \in U_b(\mathbf{C})$, the curves $(\mathcal{A}_1)_x$ and $(\mathcal{A}_2)_x$ are exactly the two nonexceptional irreducible components of C_x.

Proof. During the proof we repeatedly restrict all schemes to a nonempty open of $\mathcal{M}$ and retain the same notation; the final open will be denoted $\mathcal{M}^\circ$. By Proposition 4.12, the geometric generic fibre of $q : \mathcal{I} \to \mathcal{M}$ is reduced and has exactly two geometrically integral components, both defined over $k(\mathcal{M})$. Let $\mathcal{A}_1, \mathcal{A}_2 \subset \mathcal{I}$ be their closures. The involution exchanges them because it exchanges their generic fibres.

The morphism q is projective and of finite presentation. Apply generic flatness to q and to $\mathcal{A}_i \to \mathcal{M}$. Geometric reducedness of the fibres of q and geometric integrality of the fibres of each $\mathcal{A}_i$ then hold after restricting to a nonempty open of $\mathcal{M}$; this is the standard spreading of these properties in a flat family (Stacks Project, Tags 052A, 0578, and 0559). This proves the flatness and fibre assertions in (i), as well as the first assertion of (ii).

Since q is projective and the $\mathcal{A}_i$ are flat, their closed immersions define two sections of the relative Hilbert scheme $\mathrm{Hilb}_{\mathcal{I}/\mathcal{M}}$. Its diagonal is closed. The equalizer of the two sections is therefore closed and omits the generic point. Removing it makes $(\mathcal{A}_1)_x$ and $(\mathcal{A}_2)_x$ distinct for every geometric x.

Give $\mathcal{A} = \mathcal{A}_1 \cup \mathcal{A}_2$ its scheme-theoretic union structure. Let $\mathcal{J}$ be the ideal of $\mathcal{A} \cap \mathcal{I}^{\mathrm{ne}}$ in $\mathcal{I}^{\mathrm{ne}}$. The geometric generic fibre is reduced and consists exactly of the two generic components, so $\mathcal{J}$ vanishes there. The image of $\mathrm{Supp}(\mathcal{J})$ in the noetherian integral scheme $\mathcal{M}$ is constructible by Chevalley's theorem and does not contain its generic point. Its closure is therefore proper. Remove this closure. Then $\mathcal{J} = 0$, and

$$\mathcal{I}^{\mathrm{ne}} = \mathcal{A} \cap \mathcal{I}^{\mathrm{ne}}$$

over the remaining open. The fibrewise reducedness, the distinctness and geometric integrality of the two curve fibres, and this scheme-theoretic equality prove (ii) and (iii).

Denote the remaining open by $\mathcal{M}^\circ$. The morphism $\mathcal{M}^\circ \to \mathcal{B}$ is open because it is the restriction of the smooth morphism $\mathcal{M} \to \mathcal{B}$ (Stacks Project, Tag 056G). Hence its image $\mathcal{B}^\circ$ is nonempty and open. For $b \in \mathcal{B}^\circ(\mathbf{C})$, the fibre U_b is a nonempty open of the integral surface $(X_b)_{\mathrm{reg}}$; it is therefore dense, irreducible, and connected. The fibrewise conclusions (i)–(iii) give the final assertion. □

6. Pair convolution detects IC_X

Fix $b \in \mathcal{B}^\circ(\mathbf{C})$, put $(P, X) = (P_b, X_b)$, and let $U = U_b$ as in Theorem 5.1. Let $\rho : S \to X$ be the minimal resolution and let $j : S \to P$ be its composition with the inclusion. Put

$$K = \mathrm{IC}_X, \qquad \widetilde{K} = Rj_*\mathbf{Q}_S[2] = K \oplus \delta_0.$$

Consider

$$\widetilde{m} : S \times S \longrightarrow P, \qquad (p, q) \longmapsto j(p) + j(q),$$

and let $\sigma(p, q) = (q, p)$. We use the minus sign to denote the anti-invariant direct summand under σ.

Lemma 6.1 (Top cohomology on the split-pair locus)**.** *After replacing U by a dense open subset, put $U^+ = -U$. For $t = -x \in U^+$, the fibre $F_t = \widetilde{m}^{-1}(t)$ has exactly four one-dimensional irreducible components in its reduction: the strict transforms of $(\mathcal{A}_1)_x$ and $(\mathcal{A}_2)_x$ and the two structural components*

$$E_{\mathrm{exc}} \times \{s_t\}, \qquad \{s_t\} \times E_{\mathrm{exc}}.$$

Here s_t is the unique point of S above t. The involution σ exchanges the first two components and exchanges the last two. Consequently

$$\dim \mathcal{H}^{-2}(\widetilde{K} * \widetilde{K})_{-,t} = 2, \qquad \dim \mathcal{H}^{-2}(K * K)_{-,t} = 1.$$

After a further shrinking of U, the latter spaces form the constant rank-one local system on U^+.

Proof. Fix $t \in U^+$, write $x = -t \in U$, and let $E_{\mathrm{exc}} = \rho^{-1}(0)$. By definition, C_x parametrizes ordered pairs whose sum is $-x = t$. Hence the fibre $F_t = \widetilde{m}^{-1}(t)$ has precisely four irreducible curve components: the strict transforms of $(\mathcal{A}_1)_x, (\mathcal{A}_2)_x$ and the two structural components

$$C_{1,t} = E_{\mathrm{exc}} \times \{s_t\}, \qquad C_{2,t} = \{s_t\} \times E_{\mathrm{exc}},$$

where s_t is the unique point above t. Indeed, away from E_{exc} the resolution is an isomorphism, while a pair with one entry on E_{exc} must have the other entry over t. Thus there are no further top-dimensional components. The involution σ exchanges $(\mathcal{A}_1)_x$ with $(\mathcal{A}_2)_x$ and $C_{1,t}$ with $C_{2,t}$.

Proper base change identifies

$$\mathcal{H}^{-2}(\widetilde{K} * \widetilde{K})_t = H^2(F_t, \mathbf{Q}).$$

The top cohomology of a proper complex curve is freely generated by the fundamental classes of the irreducible components of its reduction. Thus $H^2(F_t, \mathbf{Q})_-$ has the explicit basis

$$[\,(\mathcal{A}_1)_x\,] - [\,(\mathcal{A}_2)_x\,], \qquad [C_{1,t}] - [C_{2,t}].$$

Here and below the same notation is used for a curve and its strict transform in F_t. Holomorphic transposition preserves complex orientation, so

$$\dim \mathcal{H}^{-2}(\widetilde{K} * \widetilde{K})_{-,t} = 2. \tag{6.1}$$

Expand the decomposition $\widetilde{K} = K \oplus \delta_0$:

$$\widetilde{K} * \widetilde{K} = (K * K) \oplus (K * \delta_0) \oplus (\delta_0 * K) \oplus (\delta_0 * \delta_0).$$

At $t \neq 0$, the last term has zero stalk. The two middle terms are both canonically K, and transposition exchanges them. Since t is a smooth point of the surface X, their anti-invariant stalk in degree -2 is the line generated by $[C_{1,t}] - [C_{2,t}]$. Taking dimensions in this direct sum decomposition gives

$$\dim \mathcal{H}^{-2}(K * K)_{-,t} = 1. \tag{6.2}$$

There is no Koszul sign: K has geometric shift [2], and $(-1)^{2\cdot 2} = 1$.

After a topological stratification and a further shrinking of U^+, proper stratified local triviality identifies the four component classes in nearby fibres. Hence the total anti-invariant local system is constant of rank two, with the two displayed component differences as a global basis. The anti-invariant cross term is its constant rank-one sub-local system generated by $[C_{1,t}] - [C_{2,t}]$. The direct-sum decomposition above identifies $\mathcal{H}^{-2}(K * K)_-$ with the quotient by this sub-local system. That quotient is constant of rank one, as asserted. No choice of a complementary component class is used. □

Lemma 6.2 (Vanishing away from the theta divisor)**.** *There is a dense open subset $W \subset P \backslash X$ such that, for every $t \in W$, the scheme $X \cap (t - X)$ is a smooth connected projective curve. The involution $p \mapsto t - p$ acts as the identity on its top cohomology. In particular,*

$$\mathcal{H}^{-2}(K * K)_-|_W = 0.$$

Proof. The sum map $m|_{X\times X} : X \times X \to P$ is surjective because two translates of the ample divisor X always meet. Moreover,

$$m\big((\{0\} \times X) \cup (X \times \{0\})\big) = X.$$

Hence, for $t \notin X$, the fibre lies entirely in $X_{\mathrm{reg}} \times X_{\mathrm{reg}}$. Generic smoothness for the restricted sum map therefore gives smoothness of the general fibre. Since $t \notin X$ and $\mathrm{Stab}_P(X) = \{0\}$, the translate $t - X$ does not contain X; its restriction to the integral Cohen–Macaulay surface X is therefore an effective ample Cartier divisor. Hartshorne's connectedness theorem, applied on X, shows that this divisor, namely $X \cap (t - X)$, is connected [10, Corollary III.7.9]. The involution $p \mapsto t - p$ preserves its complex orientation and hence acts by $+1$ on its one-dimensional top cohomology. Therefore

$$\mathcal{H}^{-2}(K * K)_- = 0 \quad \text{on a dense open subset of } P. \tag{6.3}$$

□

Lemma 6.3 (Extraction of the divisor-supported summand)**.** *In the decomposition-theorem splitting of $(K * K)_-$, the constant rank-one local system of Lemma 6.1 is the restriction of an unshifted* IC_X *summand. It cannot arise from a shifted simple summand with support P, or from a summand with any other support.*

Proof. The complex $(K*K)_-$ is a direct summand of the projective direct image of a pure polarizable Hodge module. By the decomposition theorem and semisimplicity [1, 21], it is a direct sum of shifted simple intersection complexes. A summand with proper support containing the generic point of U^+ must have support X. Such a simple summand has the form $\mathrm{IC}_X(\mathcal{V})[r]$, where $\mathcal{V}$ is an irreducible local system on a dense smooth open $X_{\mathcal{V}} \subset X_{\mathrm{reg}}$. Its restriction to $U^+ \cap X_{\mathcal{V}}$ is $\mathcal{V}[2+r]$, so contribution to ordinary degree -2 forces $r = 0$.

A full-support summand cannot account for the line in (6.2). Choose a general point of U^+ outside the remaining singular strata of the full-support summands, take a normal analytic disk Δ to X, and write $j_\Delta : \Delta^* \hookrightarrow \Delta$. Locally, a simple perverse intersection complex with support P has the form

$$\mathcal{L}[2] \boxtimes (j_{\Delta,!*}\mathcal{W}[1])$$

for local systems $\mathcal{L}$ and $\mathcal{W}$. Writing $i_0 : \{0\} \hookrightarrow \Delta$, the defining no-quotient property of intermediate extension gives

$$\mathcal{H}^0\big(i_0^* j_{\Delta,!*}\mathcal{W}[1]\big) = 0.$$

Since a perverse sheaf on a disk has ordinary stalk amplitude $[-1, 0]$, $i_0^* j_{\Delta,!*}\mathcal{W}[1]$ is concentrated in degree -1 when nonzero. Thus an unshifted full-support summand can contribute there only in ordinary degree -3. Contribution in degree -2 requires the perverse-degree-one shift $[-1]$. But on the dense smooth open of P that same summand is a nonzero local system shifted by $[3][-1]$ and therefore contributes in degree -2, contrary to (6.3). Supports of dimension at most one do not meet a generic point of U^+, and a different irreducible divisor cannot contain U^+.

The local system $\mathcal{V}$ has rank one because its contribution to (6.2) has rank one. Replace U^+ by $U^+ \cap X_{\mathcal{V}}$. Its restriction there is the constant local system found in Lemma 6.1. The inclusion $U^+ \hookrightarrow X_{\mathcal{V}}$ induces a surjection on fundamental groups because both are smooth irreducible complex varieties and U^+ is a nonempty Zariski-open subset. The monodromy of $\mathcal{V}$ is therefore trivial, so the summand is IC_X itself. □

Proposition 6.4 (Alternating support detector)**.** *In* $\mathrm{Perv}(P, \mathbf{Q})$*,* $K = \mathrm{IC}_X$ *is a direct summand of* ${}^pH^0((K * K)_-)$*. After tensoring with* $\mathbf{C}$*, its image in* $\overline{\mathrm{Perv}}(P)$ *is nonzero and is a direct summand of* $\mathrm{Alt}^2(K)$*.*

Proof. The first assertion is Lemma 6.3. Apply the quotient functor to $K \otimes_{\mathbf{Q}} \mathbf{C}$. It is perverse t-exact, convolution is t-exact in $\overline{D}(P)$, and all nonzero perverse-cohomological degrees of $K * K$ map to zero. Hence the image of ${}^pH^0((K * K)_-)$ is the anti-invariant summand of the tensor square of the image of K. Because the support dimension is even, this is $\mathrm{Alt}^2(K)$ under the fibre functor. The exact quotient functor preserves the direct summand supplied by Lemma 6.3; that summand is nonzero because $\chi(K) = 7$. □

7. Identification of the full Tannaka group

Let

$$G = G(K), \qquad V = \omega(K).$$

By (3.6), $\dim V = 7$. The symmetry (3.2), Verdier self-duality, and the definition of the rigid dual give

$$K^\vee = (-1_P)^* D(K) \simeq K.$$

Thus evaluation is a perfect pairing $K * K \to \delta_0$. The chosen symmetry is centred at the origin, so no translated skyscraper occurs, and $\dim X = 2$ makes the pairing symmetric. Therefore the full group embeds faithfully as

$$G \hookrightarrow O(V). \tag{7.1}$$

The simple perverse sheaf K is nondivisible: a translation fixing it must fix its support, and (3.7) then makes the translation zero. By [11, Proposition 3.3],

$$V|_{G^0} \quad \text{is irreducible.} \tag{7.2}$$

This result does not require the ambient abelian variety to be simple.

Choose $\alpha \in \Omega$ as in Proposition 3.7 and use the associated generic-twist fibre functor on the tensor category generated by K. Since any two fibre functors over $\mathbf{C}$ are noncanonically isomorphic, we identify this one with ω. Let $G_P = G$ and let G_H be the Tannaka group of the pure Hodge module IC_X^H, acting faithfully on the same complex vector space V. By [14, Corollary 3.2], the Hodge decomposition is defined by a cocharacter $h : \mathbf{G}_m^2 \to G_H$. The comparison theorem [14, Corollary 2.5] gives

$$G_P \triangleleft G_H, \qquad [G_P^0, G_P^0] = [G_H^0, G_H^0] =: G^*.$$

Since $G_P^0 \subset G_H^0$, equation (7.2) implies that $V|_{G_H^0}$ is irreducible. Thus the connected centre Z_H^0 acts by scalar matrices. As G_H^0 is reductive, multiplication $Z_H^0 \times G^* \to G_H^0$ is surjective with finite kernel.

Put $\mu(z) = h(z, z^{-1})$. The source of h is connected, so $\mu(\mathbf{G}_m) \subset G_H^0$. By Proposition 3.7,

$$\det \mu(z) = z^{(-2)\cdot 2 + 0\cdot 3 + 2\cdot 2} = 1.$$

The determinant is trivial on the semisimple group G^* and therefore descends to a character

$$\overline{\det} : G_H^0 / G^* \longrightarrow \mathbf{G}_m.$$

The quotient G_H^0/G^* is the image of Z_H^0. On Z_H^0 the determinant is $c \mapsto c^7$; since the faithful scalar action embeds Z_H^0 into $\mathbf{G}_m$, the induced character $\overline{\det}$ has finite kernel. Hence the image of the connected group $\mathbf{G}_m$ under μ in G_H^0/G^* is both connected and finite, and is therefore trivial. Consequently

$$\mu(\mathbf{G}_m) \subset G^*,$$

the common Hodge and perverse derived group. Its weights on V are

$$-2, 0, 2 \quad \text{with multiplicities } 2, 3, 2. \tag{7.3}$$

Applying the fibre functor to Proposition 6.4 gives

$$\mathrm{Hom}_G(V, \Lambda^2 V) \neq 0. \tag{7.4}$$

Proposition 7.1. *There is an isomorphism of pairs*

$$(G^0, V|_{G^0}) \simeq (G_2, V_7).$$

Proof. By (7.1) and (7.2), the connected centre of G^0 acts by orthogonal scalars and is therefore trivial. Thus G^0 is semisimple. An irreducible representation of a product of semisimple groups is an external tensor product. Since 7 is prime and the action is faithful, G^0 has one simple factor.

The classification of irreducible self-dual seven-dimensional representations leaves exactly

$$SO_7 \text{ on its standard module,} \qquad G_2 \text{ on its standard module,}$$
$$PGL_2 \text{ on } \operatorname{Sym}^6(\mathbf{C}^2).$$

Passing to the simply connected cover and consulting the low-dimensional highest weights gives, before imposing self-duality, precisely

$$A_1 : 6\omega_1, \qquad A_6 : \omega_1, \omega_6, \qquad B_3 : \omega_1, \qquad G_2 : \omega_1$$

[5, 7]. The two A_6 modules are dual to each other and are not self-dual; the remaining simple types have no irreducible module of dimension seven. Faithfulness then gives the three group forms displayed above. For SO_7, the exterior square of the standard module is the irreducible adjoint module, so $\operatorname{Hom}_{SO_7}(V, \Lambda^2 V) = 0$, contrary to (7.4). For a nontrivial cocharacter of PGL_2, the weights on $\operatorname{Sym}^6(\mathbf{C}^2)$ are seven distinct numbers. For a primitive cocharacter they are

$$-3, -2, -1, 0, 1, 2, 3,$$

and every other nonzero cocharacter multiplies them by a nonzero integer. The trivial cocharacter has one weight of multiplicity seven. Neither possibility agrees with (7.3). Hence $G^0 = G_2$. $\square$

Corollary 7.2 (The full Tannaka group)**.** *There is an isomorphism of pairs*

$$(G, V) \simeq (G_2, V_7).$$

In particular, G is connected; this identifies the full Tannaka group, not only its identity component.

Proof. By Proposition 7.1, $G^0 \simeq G_2$ on V_7. Since $G^0 \triangleleft G \subset O(V)$,

$$G \subset N_{O(V)}(G_2) = G_2 \times \{\pm I\}.$$

Indeed, conjugation gives a homomorphism from the normalizer to $\operatorname{Aut}(G_2)$. Since $\operatorname{Out}(G_2) = 1$, multiplication by a suitable element of G_2 puts every normalizing element in the centralizer. Schur's lemma makes that centralizer scalar, and its intersection with $O(V)$ is $\{\pm I\}$. If $G \neq G^0$, choose $g \in G \cap (-G_2)$ and write $g = (-I)h$ with $h \in G_2 = G^0$. Then $-I = gh^{-1} \in G$. But $-I$ acts by -1 on V and by $+1$ on $\Lambda^2 V$, contradicting (7.4). Hence $G = G^0 = G_2$. $\square$

Proof of Theorem 1.1. Take $\mathcal{B}^\circ$ to be the nonempty open subset constructed in Theorem 5.1, and fix $b \in \mathcal{B}^\circ(\mathbf{C})$. The geometric assertions about (P_b, X_b) follow from Proposition 3.3. For this fixed b, the common generic-vanishing locus Ω_b used in Proposition 3.7 is nonempty; choose any $\alpha_b \in \Omega_b$ and its unitary local system $\mathcal{L}_{\alpha_b}$. That proposition then gives the Hodge cocharacter multiplicities $(2, 3, 2)$. The open set U_b and the two pair components supplied by Theorem 5.1 give, through Proposition 6.4, that IC_{X_b} is a direct summand of $\operatorname{Alt}^2(\mathrm{IC}_{X_b})$. Finally, Corollary 7.2 identifies the resulting group and faithful module with (G_2, V_7). Since b was arbitrary in $\mathcal{B}^\circ(\mathbf{C})$, all conclusions hold with the quantifier stated in the theorem. $\square$

Acknowledgements and disclosure

The author thanks Daniel Litt for encouraging a more checkable exposition. OpenAI language-model tools assisted with drafting, symbolic-code review, and proof auditing. The computer-assisted assertions used in the proof are exact, are delimited in the Introduction and Appendix, and can be independently reproduced from the accompanying source files.

Appendix A. Computer-assisted exact norm certificate

The calculations used above fall into three logically separate groups.

Derivation of the determinant formula.
`code/g2_mu6_source_provenance_verify.py` derives (4.7) from the cyclic norm and checks the degrees, homogeneity, and multiplicity of $u - v$.

Verification of the norm identity.
Two exact-arithmetic scripts verify (4.9) by algebraically different reductions of the same generated sextic and coefficient formulas:

- `code/g2_mu6_lowmem_certificate_fast.py`;
- `code/g2_mu6_direct_numerator_verify.py`.

The first works in the exact fraction field; the second uses a separate numerator-clearing reduction.

Finite-field nonemptiness checks.
The provenance verifier applies Rabin's criterion and checks a smooth rational point at the specialization $(a, b, c) = (1, 2, 3)$, establishing geometric integrality of the residual-octic fibre used in Lemma 4.3. The Singular wrapper independently repeats that factorization and smooth-point check. Exact finite-field checks in `code/g2_mu6_nonsimple_norm_conic_Q.py` establish the simultaneous nonvanishing used in Lemma 4.9. The first two groups certify identities over $\mathbf{Q}$; the finite-field witnesses certify only the nonemptiness of specified open conditions. The generic and uniform conclusions are supplied by the geometric arguments in the main text.

On $e_1 = v = 1$, write

$$F = z^6 + c_5 z^5 + c_4 z^4 + c_3 z^3 + c_2 z^2 + c_1 z + c_0$$

and put $k = 3\Delta$. The leading coefficient of the unnormalized sextic $\widetilde{F}$ in Proposition 4.5 is

$$\operatorname{lc}_z(\widetilde{F}) = \frac{L_0}{9 e_2 \lambda^2 (e_2 - e_3)},$$

where the normalization is fixed by

$$\begin{aligned} L_0 = {} & e_2^6 \lambda^2 (u^2 + u + 1) - 3 e_2^4 e_3 \lambda^2 (u - 1)(2u + 1) \\ & + 3 e_2^3 e_3 \lambda (1 - \lambda)(u + 1) + 9 e_2^2 e_3^2 \lambda^2 (u - 1)^2 \\ & + 9 e_2 e_3^2 \lambda (\lambda - 1)(u - 1) + 3 e_3^2 (\lambda - 1)^2. \end{aligned}$$

Set

$$A = z^3 + a_2 z^2 + a_1 z + a_0, \qquad B = b_2 z^2 + b_1 z + b_0,$$

where

$$b_2 = \frac{e_2\lambda}{2L_0}(2e_2^3\lambda u + e_2^3\lambda - 6e_2e_3\lambda u + 6e_2e_3\lambda - 3e_3\lambda + 3e_3),$$
$$b_1 = -\frac{\lambda}{2L_0}(e_2^3\lambda u + 2e_2^3\lambda + 3e_2^2e_3\lambda u^2 - 6e_2^2e_3\lambda u + 3e_2^2e_3\lambda - 3e_2e_3\lambda u + 3e_2e_3\lambda - 3e_3\lambda + 3e_3),$$
$$b_0 = \frac{\lambda(u-1)}{2L_0}(-e_2^3\lambda + 3e_2e_3\lambda u - 3e_2e_3\lambda + 2e_3\lambda - 2e_3),$$

and

$$a_2 = \frac{c_5}{2}, \qquad a_1 = \frac{c_4 - a_2^2 - kb_2^2}{2}, \qquad a_0 = \frac{c_3}{2} - a_2a_1 - kb_2b_1.$$

The leading coefficients agree because F and A are monic. The identity $F = A^2 + kB^2$ is therefore equivalent to the remaining six coefficient equalities:

$$\begin{aligned}
c_5 &= 2a_2,\\
c_4 &= a_2^2 + 2a_1 + kb_2^2,\\
c_3 &= 2a_0 + 2a_2a_1 + 2kb_2b_1,\\
c_2 &= a_1^2 + 2a_2a_0 + k(b_1^2 + 2b_2b_0),\\
c_1 &= 2a_1a_0 + 2kb_1b_0,\\
c_0 &= a_0^2 + kb_0^2.
\end{aligned}$$

The first script reduces the six left-minus-right expressions to zero in the exact fraction field

$$\mathbf{Q}(\lambda, e_2, e_3, u).$$

The second clears the common denominator $4\lambda^2L_0^2$ from $F - A^2 - kB^2$ and obtains the zero polynomial. These are exact symbolic calculations, not numerical sampling.

Starting directly from the cyclic norm, the provenance calculation gives

$$\texttt{DETERMINANT\_DEGREE} = 9, \qquad \texttt{STRUCTURAL\_FACTOR\_MULTIPLICITY} = 1,$$
$$\texttt{RESIDUAL\_FACTOR\_DEGREE} = 8, \qquad \texttt{RESIDUAL\_FACTOR\_HOMOGENEOUS} = \texttt{True},$$
$$\texttt{RABIN\_Q4\_GCD} = 1, \qquad \texttt{RABIN\_Q8\_REMAINDER} = 0.$$

At $(a, b, c) = (1, 2, 3)$ modulo 13, the geometric-integrality calculation has one nonconstant factor of degree eight and exponent one. On the primitive numerator $35640H$, the affine point $(u, v) = (0, 2)$ gives

$$\texttt{POINT\_VALUE} = 0, \qquad \texttt{POINT\_GRADIENT} = (-2, 0).$$

The second sample is the curve $y^2 = x^3 - 64x + 64$ with $(a, b, c) = (1, 2, -2)$ modulo 13. It uses a kernel basis (k_0, k_1, k_2) related to the universal basis by

$$k_0 = s_0, \qquad k_1 = 5s_0 + 4s_1 - s_2, \qquad k_2 = s_0 + s_1.$$

The branch discriminant and fixed rank-witness minors for the evaluation maps on $H^0(C, 3\kappa)$ and $H^0(C, 2\kappa)$ are

$$\texttt{BRANCH\_DISCRIMINANT\_MOD13} = 8,$$
$$\texttt{EVAL\_3K\_MINOR\_COLUMNS\_0\_1\_3\_MOD13} = 1,$$
$$\texttt{EVAL\_2K\_MINOR\_COLUMNS\_0\_1\_2\_MOD13} = 1.$$

Thus the elliptic curve remains smooth and both evaluation maps retain rank three after reduction modulo 13. The script first identifies the internal point $(1 : 0 : 9)$

with the universal point $(10 : 9 : 0)$ and verifies that the corresponding conic has coefficient vector $(4, 3, 1, 2, 3, 12)$ modulo 13. Direct exact arithmetic gives the following nonzero residues:

$$\begin{aligned} \texttt{FIXED\_POINT\_LINE\_VALUES} &= (12, 8, 6, 12, 9, 1), \\ \texttt{LINE\_SECTION\_DERIVATIVE\_RESULTANTS} &= (10, 6), \\ \texttt{LINE\_BRANCH\_RESULTANTS} &= (12, 1), \\ \texttt{LINE\_PAIR\_RESULTANT} &= 12 \end{aligned}$$

in $\mathbf{F}_{13}$. These four lines verify, in order, avoidance of the fixed divisor, transversality of the two residual line sections, avoidance of the branch divisor, and disjointness of the two residual sections. The source equations, coefficient formulas, and verification scripts are the supplementary files

- `code/g2_mu6_source_provenance_verify.py`;
- `code/g2_mu6_lowmem_certificate_fast.py`;
- `code/g2_mu6_direct_numerator_verify.py`;
- `code/g2_mu6_nonsimple_norm_conic_Q.py`;
- `code/g2_mu6_universal_special123_mod13_audit.ps1`.

The versioned generated inputs used by the first three Python verifiers are in `code/generated/`; the fourth Python script can generate its optional Singular inputs with `-write-inputs`. The provenance verifier derives the saved Singular inputs from the normalized elliptic cubic and the three sections used above, then compares the resulting formulas and complete files. The scripts use exact SymPy arithmetic and can be run with

```
python code/g2_mu6_source_provenance_verify.py
python code/g2_mu6_lowmem_certificate_fast.py
python code/g2_mu6_direct_numerator_verify.py
python code/g2_mu6_nonsimple_norm_conic_Q.py
```

With Singular installed in WSL, the finite-field factorization and smooth-point checks are reproduced by

```
$audit = "code/g2_mu6_universal_special123_mod13_audit.ps1"
powershell -NoProfile -ExecutionPolicy Bypass -File $audit
```

The complete source and versioned release record are available at https://github.com/FDmd233/litt-problem-13-g2.

References

1. A. A. Beilinson, J. Bernstein, and P. Deligne, *Faisceaux pervers*, Astérisque, vol. 100, Société Mathématique de France, Paris, 1982.
2. Christina Birkenhake and Herbert Lange, *Complex abelian varieties*, second ed., Grundlehren der mathematischen Wissenschaften, vol. 302, Springer-Verlag, Berlin, 2004.
3. Jean-Paul Brasselet, Dũng Tráng Lê, and José Seade, *Euler obstruction and indices of vector fields*, Topology **39** (2000), no. 6, 1193–1208.
4. Mark Andrea A. de Cataldo and Luca Migliorini, *The hard Lefschetz theorem and the topology of semismall maps*, Ann. Sci. École Norm. Sup. (4) **35** (2002), no. 5, 759–772.
5. E. B. Dynkin, *Semisimple subalgebras of semisimple Lie algebras*, Amer. Math. Soc. Transl. Ser. 2 **6** (1957), 111–244.
6. J. Franecki and M. Kapranov, *The Gauss map and a noncompact Riemann–Roch formula for constructible sheaves on semiabelian varieties*, Duke Math. J. **104** (2000), no. 1, 171–180.
7. William Fulton and Joe Harris, *Representation theory: A first course*, Graduate Texts in Mathematics, vol. 129, Springer-Verlag, New York, 1991.

8. Mark Green and Robert Lazarsfeld, *Deformation theory, generic vanishing theorems, and some conjectures of Enriques, Catanese and Beauville*, Invent. Math. **90** (1987), no. 2, 389–407.
9. Alexander Grothendieck and Jean Dieudonné, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas, troisième partie*, Publ. Math. Inst. Hautes Études Sci. **28** (1966), 5–255.
10. Robin Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, vol. 52, Springer-Verlag, New York, 1977.
11. Ariyan Javanpeykar, Thomas Krämer, Christian Lehn, and Marco Maculan, *The monodromy of families of subvarieties on abelian varieties*, Duke Math. J. **174** (2025), no. 6, 1045–1149.
12. Masaki Kashiwara and Pierre Schapira, *Sheaves on manifolds*, Grundlehren der mathematischen Wissenschaften, vol. 292, Springer-Verlag, Berlin, 1990.
13. George Kempf, *On the geometry of a theorem of Riemann*, Ann. of Math. (2) **98** (1973), no. 1, 178–185.
14. Thomas Krämer, Christian Lehn, and Marco Maculan, *Exceptional Tannaka groups only arise from cubic threefolds*, J. Reine Angew. Math. **830** (2026), 51–99.
15. Thomas Krämer and Rainer Weissauer, *Vanishing theorems for constructible sheaves on abelian varieties*, J. Algebraic Geom. **24** (2015), no. 3, 531–568.
16. Herbert Lange and Angela Ortega, *Prym varieties of cyclic coverings*, Geom. Dedicata **150** (2011), no. 1, 391–403.
17. Henry B. Laufer, *On minimally elliptic singularities*, Amer. J. Math. **99** (1977), no. 6, 1257–1295.
18. Daniel Litt, *Problem no. 13*, Problems I Like, Accessed July 27, 2026.
19. I. G. Macdonald, *Symmetric products of an algebraic curve*, Topology **1** (1962), 319–343.
20. Henry C. Pinkham, *Simple elliptic singularities, del Pezzo surfaces and Cremona transformations*, Several Complex Variables, Proc. Sympos. Pure Math., vol. 30, American Mathematical Society, Providence, RI, 1977, pp. 69–71.
21. Morihiko Saito, *Modules de Hodge polarisables*, Publ. Res. Inst. Math. Sci. **24** (1988), no. 6, 849–995.